

MSc Applied Mathematics
Track: Data Science and Optimization

Master thesis

Exploring clustering and sparsity structures in ANCOVA models

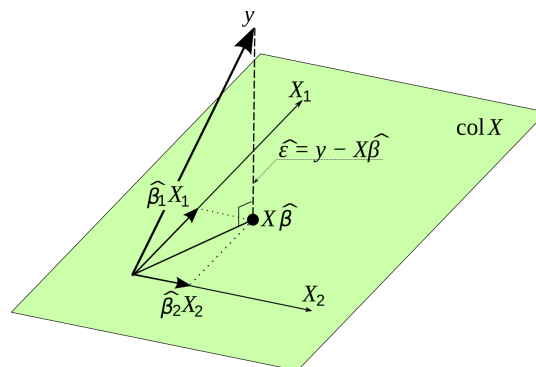
by

Christina Papoula

July, 2026

Supervisor: Eduard Belitser

Second examiner: Sandjai Bhulai



Department of Mathematics
Faculty of Sciences

Acknowledgments

First of all, I would like to thank my supervisor, Eduard, for being with me throughout this year. I am grateful that he gave me the opportunity to explore a topic connected to my own interests and the space to develop this work in my own way while also understanding the challenges and difficulties I faced along the way. I would also like to thank my partner for being so supportive, patient, and encouraging during these final and most demanding months of this master's thesis project. To my mother and dad, thank you for your unconditional love and support throughout my life. Mom, thank you for believing in me during the moments when I did not believe in myself. Dad, thank you for showing me the path towards mathematics, although I am sorry to say that after eight years of studying it, I think I have finally had enough. I would also like to thank all my friends for always being there for me and for trying to encourage and support me in every way they could. A special thanks goes to my students. During the moments when I faced difficulties with my studies, they reminded me how proud they were of their mathematics teacher of completing such a demanding degree.

Abstract

Analysis of covariance (ANCOVA) is a linear model that combines continuous and categorical explanatory variables, making it a natural setting in which two common structures can appear: sparsity and clustering. The main challenge is to identify a suitable and meaningful model structure among all possible ones. The idea of this thesis is to search over linear subspaces determined by these structures and select the one whose projection fits well while penalizing the complexity of that subspace. This can be achieved by minimizing a penalized criterion. We properly define these subspaces using the unit notion so that sparsity becomes unit sparsity, and clustering becomes clustering within units. We then quantify their complexity by a penalty called the majorant, which accounts for both the dimension of the subspace and a combinatorial term. Unit sparsity for categorical variables is shown to be an extreme case of clustering within units. The chosen models for this work are ANCOVA with one factor and q covariates without interactions, one way and two way ANOVA. For these low-dimensional settings, we motivate an adjusted majorant as a finite-sample calibration with empirical rather than theoretical support. A specific form of this majorant gives our penalized criterion a form closely related to the classical model selection criteria. The penalized criterion is studied empirically through simulations, and tuning of its parameters takes place via recovery rates and cross-validation; the penalty of the combinatorial term behaves differently for the two structures. A real-data application on the Palmer penguins dataset shows that the penalized criterion, tuned through a relaxed cross-validation rule, selects an interpretable and meaningful structure while sacrificing nothing in prediction error.

Keywords: ANCOVA, projection framework, sparsity, clustering.

Title: Exploring clustering and sparsity structures in ANCOVA models

Author: Christina Papoula, c.papoula@student.vu.nl, 2770558

Supervisor: Eduard Belitser

Second examiner: Sandjai Bhulai

Date: July, 2026

Department of Mathematics

VU University Amsterdam

de Boelelaan 1081, 1081 HV Amsterdam

<http://www.math.vu.nl/>

Contents

1. Introduction	6
1.1. Motivation	6
1.2. Notations and conventions	7
1.3. ANCOVA models	8
1.4. Combinatorial quantities	11
2. General Framework for projection structures	13
2.1. Framework setup and projection structures	13
2.2. Model selection and projection estimation	14
2.3. Conditions and majorant construction	15
2.4. ANCOVA as a particular case of the framework	16
3. Structural framework in ANCOVA models	18
3.1. Explored structures	18
3.1.1. Unit notion	18
3.1.2. Unit sparsity $I^{(S)}$	20
3.1.3. Clustering within units $I_{\mathcal{P}}$	23
3.1.4. Unified view of the two structures for multi-column units	27
3.2. Majorant constructions	28
3.2.1. ANCOVA: One factor and q covariates	29
3.2.2. Two way ANOVA	31
3.3. Motivation for adjusting the majorant	33
3.3.1. Connection to classical criteria.	35
4. Simulation study in ANCOVA models	37
4.1. Simulation setup for the one way ANOVA model	37
4.1.1. Tuning via recovery rates	38
4.1.2. Cross-validation tuning	39
4.1.3. Selection using both tuning values	41
4.2. The role of $M_{\text{true}} = 2$ and large J for the theoretical majorant	42
4.3. Extending the study to the ANCOVA model	44
4.3.1. Simulation setup	44
4.3.2. Tuning via recovery rates	44
5. Real data study in two way ANOVA models	48
5.1. The Palmer penguins dataset	48
5.1.1. Data preprocessing	49

5.2. Model set up	49
5.2.1. Preliminary classical analysis	49
5.2.2. Two way ANOVA formulation	49
5.3. Applying the projection framework	50
5.3.1. Model selection and penalty calibration	50
5.3.2. Projection estimation	54
6. Discussion	56
6.1. Contribution of this thesis and main findings	56
6.2. Limitations and possible extensions	58
6.3. The benefits of applying the general framework for projection structures to ANCOVA models	59
A. Appendix	61
A.1. Additional simulation results for section 4.1.1	62
A.2. Additional simulation results for section 4.1.2	63
A.3. Additional simulation results for section 4.2	64
A.4. Additional simulation results for section 4.3.2	65
A.5. Complete candidate model table for 6.3.1	68
A.6. Signal-to-noise ratio computations for the Palmer penguins data study .	69
A.7. Code Availability	70
Bibliography	71

1. Introduction

1.1. Motivation

Consider the linear model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \sigma\mathbf{e} \tag{1.1.1}$$

where $\sigma > 0$ (known) noise intensity and $\mathbf{e} \sim N(0, \mathbf{I}_n)$. We want to explain the response vector $\mathbf{Y} \in \mathbb{R}^n$ of continuous random observations through non-random explanatory variables x_{ij} , (with $i = 1, \dots, n$ and $j = 0, \dots, p$), stored in the design matrix $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$ and associated with unknown coefficients $\boldsymbol{\beta} = (\beta_0, \dots, \beta_p)^\top \in \mathbb{R}^{p+1}$. Once a model structure is fixed, ordinary least squares (OLS) simply projects $\mathbf{Y}$ orthogonally onto the chosen model subspace, minimizing the residual sum of squares (RSS) and is easy to compute $\hat{\boldsymbol{\beta}}$. The hard part is rather to choose a suitable and meaningful model structure among all possible ones, that is, to decide which structure to impose on $\boldsymbol{\beta}$. Two common structures are sparsity, where irrelevant explanatory variables have coefficients equal to zero, and clustering, where variables that explain the response in a similar way share one common coefficient.

Analysis of covariance (ANCOVA) is a linear model that combines continuous and categorical explanatory variables [7], making it a natural setting where both of these structures can appear [16]. For a continuous variable, the question is whether it should be included in the model at all (sparsity). For a categorical variable with distinct levels, the question is richer: one has to decide not only whether to include the factor or not, but also which of its levels should be distinguished or merged (clustering) [14, 17].

Several existing approaches already study this structural problem in linear models with continuous and categorical explanatory variables. All of these have two common points. The first is that the sparsity for continuous variables differs from the structural treatment needed for categorical variables. A categorical factor is represented by a group of coefficients, which requires a different approach from simply setting individual coefficients equal to zero. The second point is that they all perform this structure selection through the coefficients $\boldsymbol{\beta}$. Classical ANCOVA does so through inclusion tests and post-hoc pairwise comparisons [6, 7]. However, a drawback of this approach is that it can lead to overlapping groups of levels rather than to a clean clustering. Penalized regression methods provide a different route, this time by penalizing the coefficients themselves or their pairwise differences. Gertheiss and Tutz [15] developed an L_1 -penalization method on pairwise differences, where some differences are shrunk to zero, giving both sparse modeling and clustering of categories. Bondell and Reich proposed the CAS-ANOVA

method[13], which addresses the shrinkage and collapsing of levels simultaneously via a similar L_1 -penalty for pairwise differences. This work was later extended in GASH-ANOVA [14] to cover models with interactions, under the heredity-type structure. Other methods, such as DMR [16] and PDMR [18], use a two-step process: first, reducing the dimensionality using squared t-tests or the Group Lasso and then choosing the model by minimizing an information criterion. Finally, Behdin et al. [17] propose an algorithm that obtains high-quality feasible solutions via block coordinate descent.

This motivates the central idea of this thesis: to search over linear subspaces instead of individual coefficients or their differences, and to choose the best one onto which to orthogonally project $\mathbf{Y}$. This is achieved by minimizing a criterion that balances goodness of fit (RSS) against a penalty for the complexity of the subspace. In other words, we penalize its geometry rather than the coefficients. The quantity of interest then becomes the mean vector $\mathbf{X}\boldsymbol{\beta}$ rather than $\boldsymbol{\beta}$ alone, estimated by this orthogonal projection. The methodology we adopt is the general framework for projection structures developed by E.Belitser and N.Nurushev [11]. This leads to three goals: to define the linear subspace determined by the sparsity and clustering structures (Chapter 3.1); to construct the complexity penalty (Chapter 3.2); and finally to empirically study the criterion via simulations (Chapter 4) and real data (Chapter 5).

1.2. Notations and conventions

In this section, we introduce the notation and conventions used throughout the thesis, focusing on the most frequently used ones; additional conventions are defined when needed.

We write $\mathbb{N}$ for the set of natural numbers and $\mathbb{R}_+$ for the set of non-negative real numbers. The symbol $:=$ refers to equality by definition. For positive sequences a_n and b_n , $a_n \lesssim b_n$ means $a_n \leq cb_n$ for all $n \in \mathbb{N}$ with some constant $c > 0$. For a finite set A , we write $|A|$. Calligraphic uppercase letters, such as $\mathcal{I}, \mathcal{P}$, are used for collections or families of objects, such as families of sets, structures, or partitions).

Vectors and matrices, random or deterministic, are written in bold letters. They are not necessarily distinguished by lowercase or uppercase notation; the intended meaning should be clear from the context. For $n \in \mathbb{N}$, the space of real-valued vectors of length n is denoted by $\mathbb{R}^n$ and is equipped with the standard inner product $\langle x, y \rangle$ of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. The notation $\|\cdot\|^2$ is used for the square of the standard Euclidean norm in $\mathbb{R}^n$. For a vector or matrix, for example $\mathbf{X}$, its transpose is denoted by $\mathbf{X}^\top$ and its inverse by $\mathbf{X}^{-1}$. The identity matrix $n \times n$ is written as $\mathbf{I}_{n \times n}$. Consequently, $\mathbf{1}_n$ denotes the vector in $\mathbb{R}^n$ whose entries are equal to one. For any linear subspace $\mathbb{L}_I \subset \mathbb{R}^n$ we write $\mathbf{P}_I$ (instead of $\mathbf{P}_{\mathbb{L}_I}$) for the orthogonal projection onto that subspace $\mathbb{L}_I$. The multivariate normal distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ will be denoted by $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Finally, subscripts and superscripts are used to indicate the role of a symbol in its local context. Later in the thesis, clustering structures will be described through partitions. In that case, the relevant partition symbol will be used as a subscript for the structure and its family, for example $I_{\mathcal{P}} \in \mathcal{I}_{\mathcal{P}}$. Sparsity structures will be described through an active set S , and for this reason we use the superscript S . Thus, $I^{(S)} \in \mathcal{I}^{(S)}$ refers to the corresponding objects associated with the active set S .

1.3. ANCOVA models

We first define the models used in this thesis. The material is standard and can be found in standard books on linear regression and analysis of covariance; we mainly follow [6, 7, 8]. The linear model (1.1.1) serves as an umbrella for various ANCOVA models, obtained by choosing the appropriate design matrix $\mathbf{X}$ that reflects the nature of the explanatory variables. Since ANCOVA forms a broad class of models, we limit the scope of this thesis to two main models and their nested cases: ANCOVA with one factor and q covariates without interactions and two way ANOVA models.

ANCOVA model(one factor and q covariates without interactions)

Suppose that there is one factor of interest, called factor A, with J distinct levels and q continuous covariates that do not interact with the factor or with each other. From each level $i \in \{1, 2, \dots, J\}$ we sample $n_i \geq 1$ observations. The total number of observations is therefore $n = \sum_{i=1}^J n_i$. This model is then given by:

$$\omega_{A\beta} : Y_{ij} = \mu + \alpha_i + \sum_{k=1}^q \beta_k x_{ijk} + \sigma e_{ij}, \quad e_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1) \quad (1.3.1)$$

for $i = 1, \dots, J$, $j = 1, \dots, n_i$.

Here:

- Y_{ij} is the j -th observation from the level i of factor A.
- μ is the overall mean.
- α_i is the main effect of factor A at the level i .
- β_k is the common regression coefficient for the k -th covariate.
- x_{ijk} is the value of the k -th covariate for the j -th observation at the i -th level of factor A.

Remark 1.3.2. *We assume normality for the error term, as it is standard and fundamental in the classical ANCOVA framework. It ensures that the test statistics used for model comparisons follow an F -distribution. In the context of this thesis, this assumption plays a critical role in satisfying the requirements of the general projection framework, so, for our convenience, we keep it.*

Remark 1.3.3. *In the implementation of Equation (1.3.1), it is assumed that the covariates x_{ijk} have been centered around their respective grand means $(\bar{x}_{..m})$. This ensures that μ is interpreted as the mean of the response variable at the average value of the covariates [7].*

The matrix form of (1.3.1) is exactly the equation (1.1.1) for $p = J + q$.

In line with standard linear-model formulations for models containing both categorical and continuous variables, we can write the design matrix in a form that separates the factor part from the covariate part [16, 17]. Then, the design matrix can be seen as a block matrix of the form $\mathbf{X} = [\mathbf{1}_n \ \mathbf{X}_A \ \mathbf{X}_\beta]$ where:

- The first column $\mathbf{1}_n$ corresponds to the intercept, that is, μ .
- $\mathbf{X}_A \in \mathbb{R}^{n \times J}$ is the ANOVA component, that is, the design matrix corresponding to factor A. Its J columns consist of ones in the entries corresponding to $Y_{i1}, \dots, Y_{in_i}$ and zeros in all the other entries.
- $\mathbf{X}_\beta \in \mathbb{R}^{n \times q}$, is the regression component, that is, the design matrix corresponding to the covariates. Its columns contain the values of the q continuous covariates for each of the n observations.

Similarly, the parameter vector can be viewed as $\boldsymbol{\beta} = (\beta_0, \boldsymbol{\beta}_A, \boldsymbol{\beta}_\beta)^\top = (\mu, \alpha_1, \dots, \alpha_J, \beta_1, \dots, \beta_q)^\top$.

We will refer to $\omega_{A\beta}$ (1.3.1) as the **full ANCOVA model**, which means that each factor level has its own distinct effect α_i , while the covariates q explain additional variability through the regression coefficients $\beta_1, \dots, \beta_q$, which are the same across all factor levels. Although ANCOVA can encompass multiple factors or interaction terms, in this thesis, we focus on the single-factor, multi-covariate structure without interactions.

The complete ANCOVA model $\omega_{A\beta}$ (1.3.1) can be reduced to simpler (nested) models depending on the presence of explanatory variables, directly affecting the design matrix $\mathbf{X}$.

The linear regression

If there are no factor effects, i.e., $\alpha_i = \mu$ for all i , so that all factor levels have the same effect and this common contribution is absorbed into the intercept, and all q covariates are present, i.e., $\beta_k \neq 0$ for all k , then the model corresponds to the classical linear regression setting:

$$\omega_\beta : \mathbf{Y} = \mathbf{1}_n \mu + \mathbf{X}_\beta \boldsymbol{\beta}_\beta + \sigma \mathbf{e} \quad (1.3.4)$$

If the intercept and the q covariate columns are linearly independent, the design matrix $\mathbf{X} = [\mathbf{1}_n \mathbf{X}_\beta]$ has full rank, $\text{rank}(\mathbf{X}) = q + 1$.

The one-factor model(One way ANOVA)

Conversely, if the regression coefficients for the covariates are all zero and all levels of the factor have a distinct effect, the model reduces to the one way ANOVA model:

$$\omega_A : \mathbf{Y} = \mathbf{1}_n \boldsymbol{\mu} + \mathbf{X}_A \boldsymbol{\beta}_A + \sigma \mathbf{e} \quad (1.3.5)$$

We will refer to ω_A (1.3.5) as the **full one way model**, meaning that each distinct level of factor A contributes differently to the explanation of the response variable.

In ANOVA models, all the entries in the design matrix $\mathbf{X} = [\mathbf{1}_n \mathbf{X}_A]$ are 1 or 0, indicating the presence or absence of the factor effects. Some columns are linearly dependent due to the intercept column $\mathbf{1}_n$, and therefore $\text{rank}(\mathbf{X}) < J + 1$.

Remark 1.3.6. *In this work, we do not impose any additional identifiability constraints on the ANOVA models. Therefore, the equality of all level effects means that their common contribution can be adsorbed into the intercept.*

The two-factor model(Two way ANOVA)

The previous one-factor setting can be extended to accommodate two categorical factors, denoted by A and B , with J and K distinct levels, respectively. This results in $J \times K$ possible treatment combinations. For each combination (i, j) , we sample $n_{ij} \geq 1$ observations, giving in total $n = \sum_{i=1}^J \sum_{j=1}^K n_{ij}$. Note that for an observation having levels from two factors, there may also be an interaction effect between the two factors.

The **full two-factor model** is given by

$$\omega_{A \times B} : Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \sigma e_{ijk} \quad (1.3.7)$$

for $i = 1, \dots, J$, $j = 1, \dots, K$ and $k = 1, \dots, n_{ij}$, where β_j represents the main effect of factor B and γ_{ij} denotes the interaction effect between the two factors.

The matrix form of this model is exactly (1.1.1) with $p = 1 + J + K + JK$. The design matrix $\mathbf{X} \in \mathbb{R}^{n \times [1+J+K+JK]}$ consists of: the intercept column, the J and K indicator columns for each factor A and the B and JK interaction columns. It can be seen as above $\mathbf{X} = [\mathbf{1}_n \mathbf{X}_A \mathbf{X}_B \mathbf{X}_\gamma]$. The parameter vector is $\boldsymbol{\beta} = (\mu, \alpha_1 \dots \alpha_J, \beta_1 \dots \beta_K, \gamma_{11} \dots \gamma_{JK})^\top$.

As above, the full two-factor model $\omega_{A \times B}$ (1.3.7) can be reduced to simpler models that affect the design matrix $\mathbf{X}$.

Additive model

If there are no interactions between factor A and B, i.e., $\gamma_{ij} = 0$ for all (i, j) , we have a model in which only the two main factor effects are present with $\mathbf{X}_{AB} = [\mathbf{1}_n \ \mathbf{X}_A \ \mathbf{X}_B] \in \mathbb{R}^{n \times [1+J+K]}$.

$$\omega_{AB} : \mathbf{Y} = \mathbf{X}_{AB}\boldsymbol{\beta}_{AB} + \sigma\mathbf{e} \quad (1.3.8)$$

The one-factor model(One way ANOVA)

This reduction assumes that only one factor contributes to the response (i.e, $\alpha_i = \gamma_{ij} = 0$ for all (i, j)) reducing the setting to the one way ANOVA model described previously. In the case where only factor B being present, we have:

$$\omega_B : \mathbf{Y} = \mathbf{1}_n\mu + \mathbf{X}_B\boldsymbol{\beta}_B + \sigma\mathbf{e} \quad (1.3.9)$$

Null model

Finally, we state the most restrictive case of all, which implies that the response is explained solely by the overall mean μ and the random error, so no factors, interactions or covariates are present:

$$\omega_0 : \mathbf{Y} = \mathbf{1}_n\mu + \sigma\mathbf{e} \quad (1.3.10)$$

At this point, it is worth mentioning that for the two way ANOVA models considered here, we follow heredity-type structures. This means that the interaction term γ_{ij} is only included in the model when both associated main effects α_i and β_j are present [14]. This idea also appears in many practical ANOVA settings, where interaction effects describe how the effect of one factor changes between the levels of the other [7, 8].

1.4. Combinatorial quantities

Recall from the introduction that one of the structures we explore is clustering. We therefore need a way to divide the levels of a factor into non-overlapping groups. This can be done using partitions and their properties. In this section, we introduce some definitions from combinatorics [3, 4] that will be used repeatedly throughout the thesis.

Definition 1.4.1. (*Partition of a finite set*) Let V be a finite set with $|V| = J$ elements. A **partition** $\mathcal{P}$ of the set V is a collection of non-empty subsets $\mathcal{P} = \{A_1, A_2, \dots, A_M\}$, called **blocks**, such that:

$$A_i \cap A_j = \emptyset \text{ for any } i \neq j, \quad \bigcup_{i=1}^M A_i = V.$$

The collection $\mathcal{P}$ is unordered, meaning that neither the order of the blocks within $\mathcal{P}$ nor the order of the elements within each block A_i changes the partition. The number of blocks in partition $|\mathcal{P}| = M$ satisfies $1 \leq M \leq J$.

Definition 1.4.2. (Stirling numbers of the second kind) The number of ways to partition a set V with $|V| = J$ elements into M non-empty blocks is given by the **Stirling number (of the second kind)** and is denoted by $S(J, M)$.

Some useful properties of Stirling numbers of the second kind that will be used in this work are the following:

1. **Recursive formula.** For positive integers J and M such that $1 \leq M \leq J$,

$$S(J, M) = S(J - 1, M - 1) + M S(J - 1, M),$$

with boundary conditions $S(J, 1) = S(J, J) = 1$ for all $J \geq 1$.

2. **Asymptotic approximation.** For fixed M , as $J \rightarrow \infty$, the Stirling number of the second kind satisfies

$$S(J, M) \sim \frac{M^J}{M!}.$$

Definition 1.4.3. (Bell Number) The total number of possible partitions of a set with J elements is given by the J -th **Bell number**, denoted by B_J . It is defined as the sum of Stirling numbers of the second kind:

$$B_J = \sum_{M=1}^J S(J, M).$$

Example 1.4.4. For $V = \{1, 2, 3\}$, the partitions into exactly two blocks are:

$$\{1, 2\}, \{3\} \quad \{1, 3\}, \{2\} \quad \{2, 3\}, \{1\}$$

so $S(3, 2) = 3$.

Additionally, there are five unique ways to partition this set:

$$\begin{array}{c} \{1\}, \{2\}, \{3\} \\ \{1, 2\}, \{3\} \quad \{1, 3\}, \{2\} \quad \{2, 3\}, \{1\} \\ \{1, 2, 3\} \end{array}$$

Note that $B_3 = \sum_{M=1}^3 S(3, k) = 1 + 3 + 1 = 5$.

2. General Framework for projection structures

In this chapter, we provide only the necessary theoretical background, constructions, and tools of the general framework of projection structures that will be used in this thesis. For further details and the full theoretical setup, the reader is referred to [11].

2.1. Framework setup and projection structures

We use a fixed-design version of the framework. That is, the design information $\mathbf{X}$ is treated as known, and the randomness is attributed to the response vector $\mathbf{Y}$ through the noise term. Thus, suppose we observe a response vector $\mathbf{Y} \in \mathbb{R}^n$ together with fixed design information $\mathbf{X}$.

$$\mathbf{Y} \sim \mathbb{P}_{\boldsymbol{\theta}}, \text{ such that } \mathbb{E}_{\boldsymbol{\theta}}(\mathbf{Y}) = \boldsymbol{\theta}, \text{ where } \boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^n.$$

$\mathbb{P}_{\boldsymbol{\theta}}$ denotes the probability measure of $\mathbf{Y}$, and $\mathbb{E}_{\boldsymbol{\theta}}$ its corresponding expectation.

Since the design information $\mathbf{X}$ is fixed, its role is absorbed into the mean vector $\boldsymbol{\theta}$. The observation vector can therefore be expressed as

$$\mathbf{Y} = \boldsymbol{\theta} + \sigma \boldsymbol{\xi}, \text{ with } \mathbb{E}_{\boldsymbol{\theta}}(\boldsymbol{\xi}) = 0. \quad (2.1.1)$$

By letting $\sigma \boldsymbol{\xi} = \mathbf{Y} - \mathbb{E}_{\boldsymbol{\theta}}(\mathbf{Y})$, the observation vector $\mathbf{Y}$ is written as its expectation plus a zero mean noise term. In the theoretical framework, $\sigma > 0$ is treated as a known noise intensity parameter and is introduced to accommodate asymptotic settings in which one studies what happens when the amount of information grows. As observations become more informative about $\boldsymbol{\theta}$, this is reflected by a decreasing noise level, $\sigma \rightarrow 0$. In later applications of this work, σ will be fixed by construction or estimated.

The mean vector $\boldsymbol{\theta}$ belongs to a parameter space $\Theta \subseteq \mathbb{R}^n$, which in theory can be high-dimensional. Then it is assumed that $\boldsymbol{\theta}$ may belong to a lower-dimensional subspace of $\Theta \subseteq \mathbb{R}^n$ due to the presence of some structure indexed by I . A countable family $\mathcal{I}$ of all admissible structures is introduced, and each $I \in \mathcal{I}$ is represented by an appropriate linear subspace $\mathbb{L}_I \subseteq \Theta$. Hence, the associated family of linear subspaces $\{\mathbb{L}_I, I \in \mathcal{I}\}$ is obtained with statistical dimension $d_I = \dim(\mathbb{L}_I)$. This family is equipped with its orthogonal projection $\mathbf{P}_I$, which is a symmetric idempotent matrix $\mathbf{P} \in \mathbb{R}^{n \times n}$, that is,

$\mathbf{P}_I^2 = \mathbf{P}_I$ and $\mathbf{P}_I^\top = \mathbf{P}_I$ [2]. It is quantified how well a single structure I can represent the mean vector $\boldsymbol{\theta}$ by the minimal projection error:

$$\min_{I \in \mathcal{I}} \{\|\boldsymbol{\theta} - \mathbf{P}_I \boldsymbol{\theta}\|^2\} = \|\boldsymbol{\theta} - \mathbf{P}_{I^*} \boldsymbol{\theta}\|^2,$$

where I^* is a minimizer in $\mathcal{I}$. If this minimum is close to zero, then $\boldsymbol{\theta}$ is said to be approximately structured according to the family $\mathcal{I}$. If this minimum is exactly zero, then $\boldsymbol{\theta}$ is exactly structured, that is, $\boldsymbol{\theta} \in \mathbb{L}_{I^*}$.

In principle, it may occur that $\mathbb{L}_I = \mathbb{L}_{I'}$ for two different structures $I \neq I'$ in the family $\mathcal{I}$. In this case, the indexing family $\mathcal{I}$ is redundant, since several labels correspond to the same underlying subspace. Without loss of generality, one may assume that the family $\mathcal{I}$ is cleaned up by keeping only one representative for each distinct subspace. This situation may arise in clustering structures; however, in the constructions considered here, structures are indexed by partitions, so different orderings of the same blocks are not treated as distinct elements. For this reason, we do not treat redundancy separately in what follows.

2.2. Model selection and projection estimation

We now introduce the model selection procedure that will be used in this thesis. We adopt the frequentist formulation of E. Belitser and N. Nurushev [11] method, which takes the form of a penalization method. Concretely, for each candidate structure $I \in \mathcal{I}$, we take the orthogonal projection $\mathbf{P}_I$ onto its linear subspace $\mathbb{L}_I \subseteq \mathbb{R}^n$. The lack of fit of structure I to observed data $\mathbf{Y}$ is measured by $RSS_I = \|\mathbf{Y} - \mathbf{P}_I \mathbf{Y}\|^2$, while the complexity of I is quantified by a penalty $\rho(I)$.

These two ingredients give rise to the penalized criterion.

$$C_\kappa(I) = RSS_I + \kappa \sigma^2 \rho(I) \quad (2.2.1)$$

where $\sigma > 0$ (known) noise intensity and $\kappa > 0$ is a tuning parameter. This criterion balances the goodness of fit, via the residual sum of squares (RSS_I), against the complexity of the considered $\mathbb{L}_I$.

The selected structure $\hat{I}$ is then defined as

$$\hat{I} = \arg \min_{I \in \mathcal{I}} C(I). \quad (2.2.2)$$

with the corresponding estimator of the mean vector $\boldsymbol{\theta}$

$$\hat{\boldsymbol{\theta}} = \mathbf{P}_{\hat{I}} \mathbf{Y}. \quad (2.2.3)$$

Therefore, both model selection and estimation are achieved within a single projection-based procedure.

2.3. Conditions and majorant construction

At this point, one also needs a way to quantify the penalty $\rho(I)$. In the chosen framework, this role is played by a function $\rho : \mathcal{I} \mapsto \mathbb{R}_+$, called the majorant, which must satisfy a certain condition (C2). This is based on a prior condition (C1) imposed on the noise term ξ , which determined the relevant statistical dimensions d_I of the spaces $\mathbb{L}_I$.

Condition C1. *For some $\alpha > 0$ and all $I \in \mathcal{I}$,*

$$\sup_{\theta \in \Theta} \mathbb{E}_{\theta}(e^{\alpha \|\mathbf{P}_I \xi\|^2}) \leq e^{d_I},$$

where d_I denotes the statistical dimension of the linear space $\mathbb{L}_I$, and reflects the complexity of the structure I , that is, the larger d_I is, the more complex the structure is. In this thesis, we take $d_I = \dim(\mathbb{L}_I)$.

We note that condition (C1) holds for $\xi \sim \mathcal{N}(0, \mathbf{I}_n)$ with $\alpha = 0.4$ and $d_I = \dim(\mathbb{L}_I)$, irrespective of the linear spaces $\mathbb{L}_I$, $I \in \mathcal{I}$.

If $\xi \sim \mathcal{N}(0, \mathbf{I}_n)$, then $\|\mathbf{P}_I \xi\|^2 \sim \chi_{d_I}^2$. By the moment generating function of $\chi_{d_I}^2$, it follows that $\mathbb{E}(e^{t \|\mathbf{P}_I \xi\|^2}) = (1 - 2t)^{-\frac{d_I}{2}}$, for any $t < \frac{1}{2}$.

Now, we want an upper bound on this function of the form $e^{C d_I}$ for some constant $C > 0$, uniform over all relevant structures I . It holds that

$$(1 - 2t)^{-\frac{d_I}{2}} \leq e^{d_I} \Leftrightarrow (1 - 2t)^{-\frac{1}{2}} \leq e, \text{ for all } d_I \geq 1.$$

Since the function $f(t) = (1 - 2t)^{-\frac{1}{2}}$ is increasing in its domain $t < \frac{1}{2}$, it is enough to choose any t such that:

$$1 - 2t \geq e^{-2} \Leftrightarrow t \leq \frac{1 - e^{-2}}{2} \approx 0.432.$$

Taking $t = 0.4$, we derive $\mathbb{E}(e^{0.4 \|\mathbf{P}_I \xi\|^2}) \leq e^{d_I}$.

We now state condition (C2) for the majorant. Since the chosen framework is developed for high-dimensional statistics, $\rho(I)$ should be large enough to control the complexity of a potentially very rich family of structures $I \in \mathcal{I}$.

Condition C2. *For some $\nu, C_\nu > 0$, the majorant function $\rho(I)$ satisfies:*

$$\sum_{I \in \mathcal{I}} e^{-\nu \rho(I)} \leq C_\nu, \text{ and } \rho(I) \gtrsim d_I, \quad I \in \mathcal{I}.$$

where $(d_I)_{I \in \mathcal{I}}$ is from condition (C1).

For each particular structure I , we need to be able to find a majorant $\rho(I)$ that meets condition (C2) in a practical way. There are several approaches to the choice of the majorant [11, p. 12–13], but in the present thesis we use the most natural (and parsimonious) default choice. We choose a penalty large enough to account for both the statistical size of each individual subspace and the overall richness of the whole family.

$$\rho(I) = d_I + \log \ell(I), \quad \ell(I) = |\mathcal{I}_d| \quad (2.3.1)$$

with $\mathcal{I}_d = \{I \in \mathcal{I} : d_I = d\}$ so that $\ell(I) = |\mathcal{I}_d|$ is the number of structures $I \in \mathcal{I}$ of the same dimension. That is, we penalize structures of the same dimensions for being hard to distinguish. Consider this as if the family $\mathcal{I}$ is decomposed into disjoint subfamilies $\mathcal{I}_d$ such that each contains all structures I corresponding to the same dimension $d_I = d$. More precisely, by letting

$$\mathcal{D} = \{d_I : I \in \mathcal{I}\}.$$

for each $d \in \mathcal{D}$, we have $\mathcal{I}_d = \{I \in \mathcal{I} : d_I = d\}$. Figure 2.1 gives a schematic visualization of this decomposition.

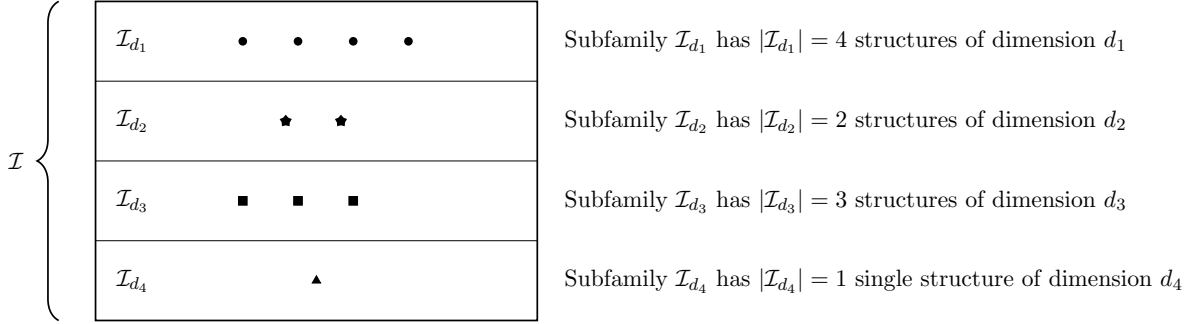


Figure 2.1.: Schematic visualization of the decomposition of the family $\mathcal{I}$ into four disjoint subfamilies $\mathcal{I}_d$. Each one contains structures having the same dimension $d_I = d$, represented by the same symbol within that subfamily. The number of structures in a subfamily is given by its cardinality $\ell(I) = |\mathcal{I}_d|$.

Condition (C2) is fulfilled for this choice; indeed, for $\nu \geq 1$,

$$\sum_{I \in \mathcal{I}} e^{-\nu \rho(I)} = \sum_{d \in \mathcal{D}} \sum_{I \in \mathcal{I}_d} e^{-\nu(d_I + \log |\mathcal{I}_{d_I}|)} \leq \sum_{d \in \mathcal{D}} e^{-\nu d_I} \leq C_\nu.$$

2.4. ANCOVA as a particular case of the framework

In this section, we explain how ANCOVA can be studied through this projection framework. Recall that we have observations $\mathbf{Y} \in \mathbb{R}^n$ that satisfy (2.1.1), where the mean vector $\boldsymbol{\theta}$ belongs to a model space $\Theta \subseteq \mathbb{R}^n$.

In the classical ANCOVA setting, $\mathbb{E}[\mathbf{Y}] = \mathbf{X}\boldsymbol{\beta}$ as in equation (1.1.1). Therefore, $\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta}$ belongs to the linear subspace $\Theta \subseteq \mathbb{R}^n$ spanned by the columns of the design matrix $\mathbf{X}$ [1]:

$$\Theta = \text{Col}(\mathbf{X}) = \{\mathbf{X}\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^{p+1}\} = \left\{ \sum_{j=0}^p \beta_j x_j : \beta_0, \dots, \beta_p \in \mathbb{R} \right\}. \quad (2.4.1)$$

Under the assumption of normality, $\mathbf{e} \sim \mathcal{N}(0, \mathbf{I}_n)$, condition (C1) is automatically satisfied, as shown in section 2.3. This, in turn, allows for a majorant construction satisfying condition (C2).

In the ANCOVA setting, each structural constraint of interest, such as sparsity or clustering, gives rise to a family of candidate structures $\mathcal{I}$. For each structure $I \in \mathcal{I}$, we construct an adjusted design matrix $\mathbf{X}_I$ by selecting or transforming the columns of $\mathbf{X}$ according to the structure I .

This defines a collection of linear subspaces $\{\mathbb{L}_I, I \in \mathcal{I}\}$, where each $\mathbb{L}_I \subseteq \text{Col}(\mathbf{X})$ is given by:

$$\mathbb{L}_I = \text{Col}(\mathbf{X}_I) = \{\mathbf{X}_I \boldsymbol{\beta}_I : \boldsymbol{\beta}_I \in \mathbb{R}^{d_I}\}. \quad (2.4.2)$$

Its statistical dimension is $d_I = \dim(\mathbb{L}_I) = \text{rank}(\mathbf{X}_I)$, that is, the maximum number of linearly independent column vectors in $\mathbf{X}$ [1]. Since each $\mathbb{L}_I$ is equipped with the orthogonal projection $\mathbf{P}_I$, projecting the observed vector $\mathbf{Y}$ onto this subspace gives the fitted mean vector under that structure I [6] :

$$\mathbf{X}_I \hat{\boldsymbol{\beta}}_I = \mathbf{P}_I \mathbf{Y}.$$

The penalized criterion (2.2.1) then applies directly to ANCOVA

$$\hat{I} = \arg \min_{I \in \mathcal{I}} C(I), \quad \mathbf{X}_{\hat{I}} \hat{\boldsymbol{\beta}}_{\hat{I}} = \mathbf{P}_{\hat{I}} \mathbf{Y}.$$

Hence, the procedure of section 2.2 applies to ANCOVA without modification, resulting in the selected structure and its mean estimate in a single step.

3. Structural framework in ANCOVA models

3.1. Explored structures

Having established that ANCOVA models naturally embed into the general projection framework, we now focus on how the structures of interest, sparsity, and clustering can be described through the columns of the design matrix. To do so, we introduce the general notion of unit. In our understanding, a unit represents a set of columns in the design matrix that conceptually belong together. A unit may consist of a single-column or several columns together.

3.1.1. Unit notion

Definition 3.1.1. (*Unit*) Let $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$ be the design matrix of the linear model (1.1.1) with vector columns $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_p$, and let $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_p)^\top$ be the vector of the corresponding coefficients.

A **unit** U is a non-empty set of column indices,

$$U \subseteq \{1, \dots, p\}, \quad U \neq \emptyset.$$

For each column index $k \in U$ we associate the vector column $\mathbf{x}_k$ of $\mathbf{X}$ and its coefficient β_k .

A unit can be either:

- **Single-column unit:** If $U = \{k\}$ for some $k \in \{1, \dots, p\}$, the unit consists of a single-column $\mathbf{x}_k$ and its coefficient β_k .
In the models considered here, this typically represents a single continuous variable (covariate).
- **Multi-column unit:** If U contains more than one index, the unit consists of several columns $\{\mathbf{x}_k : k \in U\}$ and their coefficients $\{\beta_k : k \in U\}$.
In the models considered here, this typically represents a single categorical variable (factor effect) or interaction effect between two factors.

Remark 3.1.2. *The intercept column $\mathbf{x}_0$ is a distinguished unit that corresponds to β_0 . It is always present in the model and conceptually does not belong together with any other column.*

This definition allows us to organize the design matrix $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$ of the linear model (1.1.1) by units. We partition the set of its column indices $\{0, 1, \dots, p\}$ into the intercept index $\{0\}$, (corresponding to β_0), and a set of $r \leq p$ disjoint units $U_1, \dots, U_r$. Then the design matrix can be expressed as follows:

$$\mathbf{X} = [\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_p] = [\mathbf{x}_0, \underbrace{\mathbf{x}_k : k \in U_1}_{\mathbf{X}_{U_1}}, \dots, \underbrace{\mathbf{x}_k : k \in U_j}_{\mathbf{X}_{U_j}}, \dots, \underbrace{\mathbf{x}_k : k \in U_r}_{\mathbf{X}_{U_r}}], \quad r \leq p \quad (3.1.3)$$

Each block $\mathbf{X}_{U_j}$ collects exactly the columns $\mathbf{x}_k$ whose indices belong together in the unit U_j , and each column $\mathbf{x}_k$ has an associated coefficient β_k in $\boldsymbol{\beta}$.

Example 3.1.4. *(Units in the two way ANOVA model)*

Consider the full two way ANOVA model in (1.3.7). In the corresponding linear model representation, the design matrix $\mathbf{X} \in \mathbb{R}^{n \times (1+J+K+JK)}$ in column representation is:

$$\mathbf{X} = [\mathbf{x}_\mu, \mathbf{x}_{\alpha_1}, \dots, \mathbf{x}_{\alpha_J}, \mathbf{x}_{\beta_1}, \dots, \mathbf{x}_{\beta_K}, \mathbf{x}_{\gamma_{11}}, \dots, \mathbf{x}_{\gamma_{JK}}]$$

where $\mathbf{x}_\mu$ is the intercept column and, for $i = 1, \dots, J$ and $j = 1, \dots, K$, $\mathbf{x}_{\alpha_i}$, $\mathbf{x}_{\beta_j}$ and $\mathbf{x}_{\gamma_{ij}}$ denote the indicator columns for level i of factor A , level j of factor B , and their interaction (i, j) , respectively.

The vector of the corresponding coefficients is

$$\boldsymbol{\beta} = (\mu, \alpha_1, \dots, \alpha_J, \beta_1, \dots, \beta_K, \gamma_{11}, \dots, \gamma_{JK})^\top$$

In this setting, we define the following three units, together with the intercept and their associated coefficients:

- The factor A multi-column unit: U_A
- The factor B multi-column unit: U_B
- The interaction γ multi-column unit: U_γ

$$\mathbf{X} = \left[\underbrace{\mathbf{x}_\mu}_{\text{intercept unit}}, \underbrace{\mathbf{x}_{\alpha_1}, \dots, \mathbf{x}_{\alpha_J}}_{U_A, \{\alpha_1, \dots, \alpha_J\}}, \underbrace{\mathbf{x}_{\beta_1}, \dots, \mathbf{x}_{\beta_K}}_{U_B, \{\beta_1, \dots, \beta_K\}}, \underbrace{\mathbf{x}_{\gamma_{11}}, \dots, \mathbf{x}_{\gamma_{JK}}}_{U_\gamma, \{\gamma_{ij} : i=1, \dots, J, j=1, \dots, K\}} \right]$$

Example 3.1.5. (*Units in the ANCOVA model*)

Consider the ANCOVA model with one factor and q covariates in (1.3.1). In the corresponding linear model representation, the design matrix $\mathbf{X} \in \mathbb{R}^{n \times (1+J+q)}$ in column representation is

$$\mathbf{X} = [\mathbf{x}_\mu, \mathbf{x}_{\alpha_1}, \dots, \mathbf{x}_{\alpha_J}, \mathbf{x}_{\beta_1}, \dots, \mathbf{x}_{\beta_q}],$$

where $\mathbf{x}_\mu$ and $\mathbf{x}_{\alpha_i}$, $i = 1, \dots, J$, as defined above and $\mathbf{x}_{\beta_k}$, $k = 1, \dots, q$, are the covariate columns.

The corresponding coefficient vector is

$$\boldsymbol{\beta} = (\mu, \alpha_1, \dots, \alpha_J, \beta_1, \dots, \beta_q)^\top.$$

In this setting, we define the following one factor unit and q covariate units, together with the intercept:

- The factor A multi-column unit : U_A
- The q covariate single-column units : $U_1, \dots, U_q$

$$\mathbf{X} = \left[\underbrace{\mathbf{x}_\mu}_{\text{intercept unit}}, \underbrace{\mathbf{x}_{\alpha_1}, \dots, \mathbf{x}_{\alpha_J}}_{U_A, \{\alpha_1, \dots, \alpha_J\}}, \underbrace{\mathbf{x}_{\beta_1}}_{U_1, \{\beta_1\}}, \dots, \underbrace{\mathbf{x}_{\beta_q}}_{U_q, \{\beta_q\}} \right].$$

3.1.2. Unit sparsity $I^{(S)}$

The first structure we consider is unit sparsity, denoted by $I^{(S)}$, which acts at the level of the disjoint units $U_1, \dots, U_r$, $r \leq p$, of equation (3.1.3). It simply answers the question: is the unit present or not?

For single-column units, which represent continuous covariates, unit sparsity has the usual interpretation as in linear regression. If the unit is present, it carries unique information that helps to explain the response variable. For multi-column units, which represent categorical factors or interactions, unit sparsity operates at the level of the whole group of columns. Conceptually, a present multi-column unit means that the corresponding factor (or interaction) provides explanatory information for the response; its levels are distinguished by separate effects in the model.

Unit sparsity is then indexed by an active set $S \subseteq \{1, \dots, r\}$ where $j \in S$ means that unit U_j is present and $j \notin S$ means that unit U_j is absent.

Equivalently, we can define the unit activity function $\delta : \{U_1, \dots, U_r\} \rightarrow \{0, 1\}$ as follows:

$$\delta(U_j) = \begin{cases} 1, & \text{if } j \in S \\ 0, & \text{if } j \notin S \end{cases} \quad (3.1.6)$$

A unit U_j is called present if it is active, that is, $\delta(U_j) = 1$ and absent(or sparse) if it is inactive, that is, $\delta(U_j) = 0$.

If a unit U_j is absent, all associated columns $\mathbf{x}_k$ of $\mathbf{X}$ with $k \in U_j$ are removed from the design matrix. The corresponding coefficients $\{\beta_k : k \in U_j\}$ are set to zero if U_j is a single-column unit or equal to the intercept β_0 , if U_j is a multi-column unit.

We therefore define the sparse design matrix $\mathbf{X}_{I(S)}$ as a submatrix [23] of the main design matrix $\mathbf{X}$ from (3.1.3). For each active unit $U_j, j \in S$ we keep all rows $i = 1, \dots, n$ and retain only the intercept column $\mathbf{x}_0$ and the columns belonging to active units $\{\mathbf{x}_k : k \in U_j, j \in S\}$:

$$\mathbf{X}_{I(S)} = [\mathbf{x}_0, \mathbf{X}_{U_j}, j \in S] = [\mathbf{x}_0, \dots, \mathbf{x}_k, \dots], \text{ for all } k \in U_j \text{ with } j \in S. \quad (3.1.7)$$

Therefore, the linear subspace $\mathbb{L}_{I(S)}$ determined by this unit sparsity structure is:

$$\mathbb{L}_{I(S)} = \text{Col}(\mathbf{X}_{I(S)}) \quad (3.1.8)$$

with statistical dimension $d_{I(S)} = \text{rank}(\mathbf{X}_{I(S)})$.

The associated parameter vector is then

$$\boldsymbol{\beta}_{I(S)} = (\beta_0, \dots, \beta_k, \dots)^\top, \quad \text{for all } k \in U_j \text{ with } j \in S.$$

Table 3.1 gives an overview of the possible rank bounds for the models considered in this thesis.

Model	Rank bounds of $\mathbf{X}_{I(S)}$
ANCOVA (factor A with J levels and q covariates)	$1 \leq \text{rank}(\mathbf{X}_{I(S)}) \leq J + q$
One way ANOVA (factor A with J levels)	$1 \leq \text{rank}(\mathbf{X}_{I(S)}) \leq J$
Two way ANOVA (with interactions) (factors A with J and B with K levels)	$1 \leq \text{rank}(\mathbf{X}_{I(S)}) \leq JK$

Table 3.1.: Rank bounds of the unit-sparse design matrix $\mathbf{X}_{I(S)}$

Example 3.1.9. (*Unit sparsity in the two way ANOVA model*)

Consider the full two way ANOVA model (1.3.7) with factor A having $J = 3$ levels and factor B having $K = 2$ levels. The design matrix, written in units, and the parameter vector of this model are:

$$\mathbf{X} = \left[\underbrace{\mathbf{x}_0}_{\text{intercept unit}}, \underbrace{\mathbf{x}_{\alpha_1}, \mathbf{x}_{\alpha_2}, \mathbf{x}_{\alpha_3}}_{U_A, J=3}, \underbrace{\mathbf{x}_{\beta_1}, \mathbf{x}_{\beta_2}}_{U_B, K=2}, \underbrace{\mathbf{x}_{\gamma_{11}}, \mathbf{x}_{\gamma_{12}}, \mathbf{x}_{\gamma_{21}}, \mathbf{x}_{\gamma_{22}}, \mathbf{x}_{\gamma_{31}}, \mathbf{x}_{\gamma_{32}}}_{U_\gamma, JK=6} \right] \in \mathbb{R}^{n \times (1+3+2+6)},$$

$$\boldsymbol{\beta} = \left(\mu, \underbrace{\alpha_1, \alpha_2, \alpha_3}_{U_A}, \underbrace{\beta_1, \beta_2}_{U_B}, \underbrace{(\gamma)_{11}, (\gamma)_{12}, (\gamma)_{21}, (\gamma)_{22}, (\gamma)_{31}, (\gamma)_{32}}_{U_\gamma} \right)^\top \in \mathbb{R}^{1+3+2+6}.$$

We apply unit sparsity by setting $\delta(U_A) = 0$ and $\delta(U_\gamma) = 0$, retaining only the unit U_B . Setting U_A inactive means that all factor A levels share the same effect ($\alpha_1 = \alpha_2 = \alpha_3$), which is absorbed into the intercept μ . Setting U_γ inactive removes all interaction terms. This corresponds to model (1.3.9), in which only factor B is present.

The sparse design matrix and parameter vector then become:

$$\mathbf{X}_{I(s)} = \left[\underbrace{\mathbf{x}_0}_{\text{intercept unit}}, \underbrace{\mathbf{x}_{\beta_1}, \mathbf{x}_{\beta_2}}_{U_B, K=2} \right] \in \mathbb{R}^{n \times (1+2)},$$

$$\boldsymbol{\beta}_{I(s)} = \left(\mu, \underbrace{\beta_1, \beta_2}_{U_B} \right)^\top \in \mathbb{R}^{1+2}.$$

The number of columns in the design matrix is reduced from $1 + 3 + 2 + 6 = 12$ to $1 + 2 = 3$

Example 3.1.10. (Unit sparsity in the ANCOVA model)

Consider the ANCOVA model (1.3.1) with factor A having $J = 5$ levels and $q = 3$ covariates. The design matrix, written in units, and the parameter vector of this model are as follows:

$$\mathbf{X} = \left[\underbrace{\mathbf{x}_0}_{\text{intercept unit}}, \underbrace{\mathbf{x}_{\alpha_1}, \mathbf{x}_{\alpha_2}, \mathbf{x}_{\alpha_3}, \mathbf{x}_{\alpha_4}, \mathbf{x}_{\alpha_5}}_{U_A, J=5}, \underbrace{\mathbf{x}_{\beta_1}}_{U_1}, \underbrace{\mathbf{x}_{\beta_2}}_{U_2}, \underbrace{\mathbf{x}_{\beta_3}}_{U_3} \right] \in \mathbb{R}^{n \times (1+5+3)},$$

$$\boldsymbol{\beta} = \left(\mu, \underbrace{\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5}_{U_A}, \underbrace{\beta_1}_{U_1}, \underbrace{\beta_2}_{U_2}, \underbrace{\beta_3}_{U_3} \right)^\top \in \mathbb{R}^{1+5+3}.$$

We apply unit sparsity by setting $\delta(U_2) = 0$, that is, $\beta_2 = 0$, while retaining U_A , U_1 , and U_3 as active. The column $\mathbf{x}_{\beta_2}$ is removed from the design matrix and β_2 is removed from the parameter vector.

The sparse design matrix and parameter vector become:

$$\mathbf{X}_{I(s)} = \left[\underbrace{\mathbf{x}_0}_{\text{intercept unit}}, \underbrace{\mathbf{x}_{\alpha_1}, \mathbf{x}_{\alpha_2}, \mathbf{x}_{\alpha_3}, \mathbf{x}_{\alpha_4}, \mathbf{x}_{\alpha_5}}_{U_A, J=5}, \underbrace{\mathbf{x}_{\beta_1}}_{U_1}, \underbrace{\mathbf{x}_{\beta_3}}_{U_3} \right] \in \mathbb{R}^{n \times (1+5+2)},$$

$$\boldsymbol{\beta}_{I(s)} = \left(\mu, \underbrace{\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5}_{U_A}, \underbrace{\beta_1}_{U_1}, \underbrace{\beta_3}_{U_3} \right)^\top \in \mathbb{R}^{1+5+2}.$$

The number of columns in the design matrix is reduced from $1 + 5 + 3 = 9$ to $1 + 5 + 2 = 8$.

3.1.3. Clustering within units $I_{\mathcal{P}}$

The second structure we consider is clustering within units, denoted by $I_{\mathcal{P}}$. While unit sparsity operates at the level of entire units, clustering allows for a more elaborate structure within a unit: the column indices of a unit are partitioned into blocks, called clusters, such that the coefficients associated with all indices in the same block are constrained to be equal. In this sense, clustering decides not only whether a unit is present or absent, but also how many distinct effects within that unit need to be distinguished.

In the ANCOVA models considered here, clustering within units applies only to multi-column units, that is, to categorical factors and interactions. Conceptually, this means that some levels of the factor of interest have the same effect on the response variable, so these levels can be represented by a common cluster. For continuous covariates, each unit consists of a single-column $\mathbf{x}_k$, so there are no within unit indices to merge.

Remark 3.1.11. *In principle, clustering across covariates is possible if one decides beforehand which covariates form a common unit [12]. Once such a grouping is fixed, the same construction applies. However, in this work, we do not pursue this direction and restrict clustering to categorical factors only.*

Consider the linear model (1.1.1) whose design matrix can be written as in (3.1.3).

For any unit multi-column $U_j \in \{U_1, \dots, U_r\}$, with $|U_j| = J > 1$, from (3.1.3), we take a partition

$$\mathcal{P} = \{A_1, A_2, \dots, A_M\},$$

of its index set $\{1, 2, \dots, J\}$, as defined in 1.4.1, with $1 \leq M \leq J$. Each block A_m , $m = 1, \dots, M$ ($A_1, \dots, A_M$) collects the indices of the levels that are merged into a common cluster. A unit U_j is called clustered if its original columns J $\mathbf{x}_k : k \in U_j$ are replaced by collapsed columns $M < J$, one for each block of the partition.

The partition $\mathcal{P}$ of the clustered unit imposes equality constraints on the coefficients $\beta_k : k \in U_j$ associated with U_j : for each block A_m , $m = 1, \dots, M$, all indices in the same block $k \in A_m$ share a common coefficient,

$$\beta_k = \beta_{k'} \quad \text{for all } k, k' \in A_m.$$

Thus, the original J individual coefficients $\{\beta_k : k \in U_j\}$ are replaced by M cluster coefficients, one for each block of the partition.

For the considered multi-column unit U_j , take its unit-block $\mathbf{X}_{U_j} = [\mathbf{x}_k]_{k \in U_j} \in \mathbb{R}^{n \times J}$ from (3.1.3). This column block has J indicator columns that will be replaced by the collapsed block:

$$\underbrace{\left[\sum_{k \in A_1} \mathbf{x}_k, \dots, \sum_{k \in A_M} \mathbf{x}_k \right]}_{\text{collapsed block for } U_j}$$

whose columns are the sum of the indicator columns that belong to the same block $A_1, \dots, A_M$ of the partition taken $\mathcal{P}$.

Algebraically, this transformation can take place via a matrix $\mathbf{T} \in \{0, 1\}^{J \times M}$ which maps the original column indices of $\mathbf{X}_{U_j}$ to the partitioned ones. We will call this a mapping matrix. The entries of $\mathbf{T}$ for row $j = 1, \dots, J$ and column $m = 1, \dots, M$ are defined as :

$$t_{jm} = \begin{cases} 1, & \text{if } j \in A_m \\ 0, & \text{otherwise} \end{cases}$$

Then

$$\mathbf{X}_{U_j} \mathbf{T} = \left[\sum_{k \in A_1} \mathbf{x}_k, \dots, \sum_{k \in A_M} \mathbf{x}_k \right] \in \mathbb{R}^{n \times M}$$

The clustered design matrix $\mathbf{X}_{I_{\mathcal{P}}}$ is then obtained by replacing only the block $\mathbf{X}_{U_j}$ by its collapsed one, while keeping all other blocks of the main design matrix $\mathbf{X}$ (3.1.3) unchanged

$$\mathbf{X}_{I_{\mathcal{P}}} = \left[\mathbf{x}_0, \mathbf{X}_{U_1}, \dots, \mathbf{X}_{U_j} \mathbf{T}, \dots, \mathbf{X}_{U_r} \right] = \left[\mathbf{x}_0, \mathbf{X}_{U_1}, \dots, \underbrace{\sum_{k \in A_1} \mathbf{x}_k, \dots, \sum_{k \in A_M} \mathbf{x}_k}_{\text{collapsed block for } U_j}, \dots, \mathbf{X}_{U_r} \right]. \quad (3.1.12)$$

Therefore, the linear subspace $\mathbb{L}_{I_{\mathcal{P}}}$ determined by the clustering within unit structure is:

$$\mathbb{L}_{I_{\mathcal{P}}} = \text{Col}(\mathbf{X}_{I_{\mathcal{P}}}), \quad (3.1.13)$$

with statistical dimension $d_{I_{\mathcal{P}}} = \text{rank}(\mathbf{X}_{I_{\mathcal{P}}})$.

The associated cluster parameter vector collects the intercept, the M cluster coefficients for the unit U_j , and all coefficients of the remaining units unchanged.

$$\boldsymbol{\beta}_{I_{\mathcal{P}}} = (\beta_0, \dots, \underbrace{\beta_1, \dots, \beta_M}_{M \text{ cluster effects of } U_j}, \dots)^\top. \quad (3.1.14)$$

Remark 3.1.15. *Regarding the full two way ANOVA models $\omega_{A \times B}$ (1.3.7) considered in this thesis, the following two hold :*

- *Clustering within the interaction unit U_γ (interaction factor) is induced by the partitions of the indices of the two main units (factor A and factor B) through their Cartesian product [19]. Let $\mathcal{P}_A = \{A_1, \dots, A_M\}$ be the partition of the indices of unit U_A and let $\mathcal{P}_B = \{B_1, \dots, B_N\}$ be the partition of the indices of unit U_B . Then the indices of the interaction factor (unit U_γ) are partitioned by the Cartesian product*

$$\mathcal{R} = \mathcal{P}_A \times \mathcal{P}_B = \{A_m \times B_n : m = 1, \dots, M, n = 1, \dots, N\}.$$

- The collapsed interaction unit-block of $X_{U_\gamma} \in \mathbb{R}^{n \times JK}$ is obtained via

$$X_{U_\gamma}(\mathbf{T}_A \otimes \mathbf{T}_B) \in \mathbb{R}^{n \times MN}, \quad \mathbf{T}_A \otimes \mathbf{T}_B \in \{0, 1\}^{JK \times MN},$$

where $\mathbf{T}_A \otimes \mathbf{T}_B$ is the direct product of $\mathbf{T}_A \in \{0, 1\}^{J \times M}$ and $\mathbf{T}_B \in \{0, 1\}^{K \times N}$ the mapping matrices associated with $\mathcal{P}_A$ and $\mathcal{P}_B$.

Table 3.2 gives an overview of the possible rank bounds for the models considered in this thesis.

Model	Rank bounds of $\mathbf{X}_{I_P}$
ANCOVA, factor A (J levels, M clusters), q covariates	$1 \leq \text{rank}(\mathbf{X}_{I_P}) \leq M + q$
One way ANOVA, factor A (J levels, M clusters)	$1 \leq \text{rank}(\mathbf{X}_{I_P}) \leq M$
Two way ANOVA(with interactions), factors A (J levels, M clusters), B (K levels, N clusters)	$1 \leq \text{rank}(\mathbf{X}_{I_P}) \leq MN$

Table 3.2.: Rank bounds of the clustered design matrix $\mathbf{X}_{I_P}$.

In the extreme case $M = J$ (and $N = K$ for two factors), no merging takes place, and each factor level forms its own cluster. In the other extreme case, $M = 1$ (and $N = 1$ for two factors), all levels collapse into a single cluster. These two extreme cases connect clustering within units to unit sparsity for the multi-column units, which we discuss in the next section.

Example 3.1.16. (*Clustering in the two way ANOVA model*)

Consider the full two way ANOVA model (1.3.7) with factor A having $J = 3$ levels and factor B having $K = 2$ levels as in example (3.1.9).

We now apply clustering within units.

Assume that for unit U_A (factor A), we take the partition $\mathcal{P}_A = \{\{1, 2\}, \{3\}\}$ with $M = 2$ clusters imposing the equality constraint $\alpha_1 = \alpha_2$.

For unit U_B (factor B), no clustering is applied, which in partition language means that we take $\mathcal{P}_B = \{\{1\}, \{2\}\}$ with $N = 2$ clusters.

Then, the clustering of unit U_γ (interactions) is induced by the cartesian product $\mathcal{P}_A \times \mathcal{P}_B$:

$$\mathcal{P}_A \times \mathcal{P}_B = ((\{1, 2\}, \{1\}), (\{3\}, \{1\}), (\{1, 2\}, \{2\}), (\{3\}, \{2\})),$$

with $MN = 4$ clusters imposing the constraint $\gamma_{11} = \gamma_{\gamma_{21}}$ and $\gamma_{12} = \gamma_{22}$.

The collapsed design matrix is obtained by replacing the $\mathbf{X}_{U_A}$ and $\mathbf{X}_{U_\gamma}$ blocks by their collapsed ones.

In this example, we fully demonstrate how this transformation can take place using mapping matrices. For the rest of the examples, in this work, the reader can similarly apply the same logic.

- Unit-block $\mathbf{X}_{U_A}$:

$$\mathbf{X}_{U_A} \mathbf{T}_A = \begin{bmatrix} \mathbf{x}_{\alpha_1} & \mathbf{x}_{\alpha_2} & \mathbf{x}_{\alpha_3} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- Unit-block $\mathbf{X}_{U_B}$:

$$\mathbf{X}_{U_B} \mathbf{T}_B = \begin{bmatrix} \mathbf{x}_{\beta_1} & \mathbf{x}_{\beta_2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- Unit-block $\mathbf{X}_{U_\gamma}$:

$$\mathbf{X}_{U_\gamma} (\mathbf{T}_A \otimes \mathbf{T}_B) = \begin{bmatrix} \mathbf{x}_{\gamma_{11}} & \mathbf{x}_{\gamma_{21}} & \mathbf{x}_{\gamma_{31}} & \mathbf{x}_{\gamma_{12}} & \mathbf{x}_{\gamma_{22}} & \mathbf{x}_{\gamma_{32}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Thus, the collapsed matrix will be

$$\mathbf{X}_{I_P} = \left[\underbrace{\mathbf{x}_0}_{\text{intercept unit}}, \underbrace{\mathbf{x}_{\alpha_1} + \mathbf{x}_{\alpha_2}, \mathbf{x}_{\alpha_3}}_{U_A, M=2}, \underbrace{\mathbf{x}_{\beta_1}, \mathbf{x}_{\beta_2}}_{U_B, N=2}, \underbrace{(\mathbf{x}_{\gamma_{11}} + \mathbf{x}_{\gamma_{21}}), (\mathbf{x}_{\gamma_{12}} + \mathbf{x}_{\gamma_{22}}), \mathbf{x}_{\gamma_{31}}, \mathbf{x}_{\gamma_{32}}}_{U_\gamma, MN=4} \right] \in \mathbb{R}^{n \times (1+2+2+4)}$$

with collapsed parameter vector

$$\boldsymbol{\beta}_{I_P} = \left(\mu, \underbrace{\alpha_1, \alpha_3}_{M=2}, \underbrace{\beta_1, \beta_2}_{N=2}, \underbrace{\gamma_{11}, \gamma_{12}, \gamma_{31}, \gamma_{32}}_{MN=4} \right)^\top \in \mathbb{R}^{1+2+2+4}.$$

The number of columns in the design matrix is reduced from $1 + 3 + 2 + 6 = 12$ to $1 + 2 + 2 + 4 = 9$.

Example 3.1.17. (Clustering in the ANCOVA model)

Consider the full ANCOVA model (1.3.1) with factor A having $J = 5$ levels and $q = 3$ covariates as in example (3.1.10) .

We now apply clustering within units (factor A). For unit U_A (factor A), we take the partition $\mathcal{P}_A = \{\{1, 3\}, \{2, 4\}, \{5\}\}$ with $M = 3$ clusters, imposing the equality constraints $\alpha_1 = \alpha_3, \alpha_2 = \alpha_4$.

Covariate units U_1, U_2, U_3 are single-column units and no clustering is applied to them.

The collapsed design matrix is obtained by replacing each cluster (block) of U_A by its collapsed one as demonstrated above,

$$\mathbf{X}_{I_{\mathcal{P}}} = \left[\underbrace{\mathbf{x}_0}_{\text{intercept unit}}, \underbrace{(\mathbf{x}_{\alpha_1} + \mathbf{x}_{\alpha_3}), (\mathbf{x}_{\alpha_2} + \mathbf{x}_{\alpha_4}), \mathbf{x}_{\alpha_5}}_{U_A, M=3}, \underbrace{\mathbf{x}_{\beta_1}}_{U_1}, \underbrace{\mathbf{x}_{\beta_2}}_{U_2}, \underbrace{\mathbf{x}_{\beta_3}}_{U_3} \right] \in \mathbb{R}^{n \times (1+3+3)},$$

with collapsed parameter vector

$$\boldsymbol{\beta}_{I_{\mathcal{P}}} = \left(\mu, \underbrace{\alpha_1, \alpha_2, \alpha_5}_{M=3}, \underbrace{\beta_1}_{U_1}, \underbrace{\beta_2}_{U_2}, \underbrace{\beta_3}_{U_3} \right)^{\top} \in \mathbb{R}^{1+3+3}.$$

The number of columns in the design matrix is reduced from $1+5+3 = 9$ to $1+3+3 = 7$.

3.1.4. Unified view of the two structures for multi-column units

A key observation is that clustering within units $I_{\mathcal{P}}$ is a richer structure than unit sparsity $I^{(S)}$ for multi-column units, in the sense that the latter can be recovered from the two extreme clustering cases.

Consider a present multi-column unit $U_j, j \in S$ with $|U_j| = J > 1$ with a unit block $\mathbf{X}_{U_j} = [\mathbf{x}_k]_{k \in U_j} \in \mathbb{R}^{n \times J}$. Take partition $\mathcal{P} = \{A_1, \dots, A_M\}$ of $\{1, \dots, J\}$, as defined in section (3.1.3), with $|\mathcal{P}| = M$ blocks.

The number of blocks ranges from $1 \leq M \leq J$, and the two extreme partitions are:

$$M = J : \mathcal{P}_{max} = \{\{1\}, \{2\}, \dots, \{J\}\} \text{ and } M = 1 : \mathcal{P}_{min} = \{\{1, 2, \dots, J\}\}$$

- For $M = J$ each level forms its own cluster, so all factor effects are distinct and fully present, that is, the unit is fully present $U_j, j \in S$ (or equivalently $\delta(U_j = 1)$).

Indeed, for $M = J$ the mapping matrix $\mathbf{T} \in \{0, 1\}^{J \times J} = \mathbf{I}_{J \times J}$. This gives

$$\mathbf{X}_{U_j} \mathbf{T} = \mathbf{X}_{U_j} \in \mathbb{R}^{n \times J}, j \in S$$

The clustered unit-block then coincides with the original (non-collapsed) one. Hence, $\mathbb{L}_{\mathcal{P}_{max}} = \text{Col}(\mathbf{X}_{U_j})$ is the full subspace for this unit.

- For $M = 1$ all levels collapse into one cluster, which means that no level differs from any other. The factor effect then disappears, that is $U_j, j \notin S$ (or equivalently $\delta(U_j = 0)$).

Indeed, for $M = 1$ the mapping matrix is $\mathbf{T} \in \{0, 1\}^{J \times 1} = \mathbf{1}_J$. This gives

$$\mathbf{X}_{U_j} \mathbf{T} = \mathbf{1}_n$$

The clustered unit-block collapses into the intercept column of the design matrix $\mathbf{X}$. Therefore, $\mathbb{L}_{\mathcal{P}_{min}} = \text{Col}(\mathbf{1}_n)$ the most collapsed subspace of all.

Table 3.3 gives an overview of all possible extreme partitions with the corresponding design matrices and models considered in this thesis.

Model	Extreme partition	Design matrix	Recovered model
ANCOVA (factor A , q covariates)	$M = 1$	$[\mathbf{1}_n \ \mathbf{X}_\beta] \in \mathbb{R}^{n \times (1+q)}$	ω_β (1.3.4)
	$M = J$	$[\mathbf{1}_n \ \mathbf{X}_A \ \mathbf{X}_\beta] \in \mathbb{R}^{n \times (1+J+q)}$	$\omega_{A\beta}$ (1.3.1)
One way ANOVA	$M = 1$	$\mathbf{1}_n \in \mathbb{R}^{n \times 1}$	ω_0 (1.3.10)
	$M = J$	$[\mathbf{1}_n \ \mathbf{X}_A] \in \mathbb{R}^{n \times (1+J)}$	ω_A (1.3.5)
Two way ANOVA	$M = 1, N = 1$	$\mathbf{1}_n \in \mathbb{R}^{n \times 1}$	ω_0 (1.3.10)
	$M = J, N = 1$	$[\mathbf{1}_n \ \mathbf{X}_A] \in \mathbb{R}^{n \times (1+J)}$	ω_A (1.3.5)
	$M = 1, N = K$	$[\mathbf{1}_n \ \mathbf{X}_B] \in \mathbb{R}^{n \times (1+K)}$	ω_B (1.3.9)
		$[\mathbf{1}_n \ \mathbf{X}_A \ \mathbf{X}_B] \in \mathbb{R}^{n \times (1+J+K)}$	ω_{AB} (1.3.8)
	$M = J, N = K$	$[\mathbf{1}_n \ \mathbf{X}_A \ \mathbf{X}_B \ \mathbf{X}_\gamma] \in \mathbb{R}^{n \times (1+J+K+JK)}$	or $\omega_{A \times B}$ (1.3.7)

Table 3.3.: Recovery of this thesis nested models at the extreme clustering cases with $M = |\mathcal{P}_A|$ and $N = |\mathcal{P}_B|$ denote the number of clusters for factor A and factor B respectively. Intermediate values $1 < M < J$ and $1 < N < K$ correspond to genuine clustering structures. Interaction in the two way model requires both $M > 1$ and $N > 1$.

Remark 3.1.18. *In the ANCOVA setting, the table focuses on the extreme clustering cases for the categorical factor only, while keeping all q covariates present, as in (1.3.1). In practice, not all q covariates need to be active; therefore, depending on which covariates are kept, additional nested models can arise between ω_β (1.3.4) and $\omega_{A\beta}$ (1.3.1)*

3.2. Majorant constructions

Having defined the structures of interest with their associated linear subspaces and their statistical dimensions, we can now construct explicit majorants using equation (2.3.1) for the models considered here.

3.2.1. ANCOVA: One factor and q covariates

Recall the ANCOVA model (1.3.1) with design matrix $\mathbf{X} = [\mathbf{1}_n \ \mathbf{X}_A \ \mathbf{X}_\beta] \in \mathbb{R}^{n \times (1+J+q)}$.

Following equation (3.1.3), we partition $\mathbf{X}$ into disjoint units, as follows

$$\mathbf{X} = \left[\underbrace{\mathbf{x}_\mu}_{\text{intercept unit}}, \underbrace{\mathbf{x}_{\alpha_1}, \dots, \mathbf{x}_{\alpha_J}}_{U_A}, \underbrace{\mathbf{x}_{\beta_1}}_{U_1}, \dots, \underbrace{\mathbf{x}_{\beta_q}}_{U_q} \right].$$

This consists of the intercept unit, a multi-column unit $U_A = \{1, \dots, J\}$ associated with factor A, and q single-column units $U_1, \dots, U_q$ associated with the covariates.

In this setting, we consider two independent structures: clustering within the unit U_A , $I_{\mathcal{P}_A}$, and unit sparsity of single-column units $U_1, \dots, U_q$, $I^{(S)}$. Each structure gives rise to a separate family and thus to a corresponding majorant. We denote these by $\mathcal{I}_{\mathcal{P}_A}$ and $\rho(I_{\mathcal{P}_A})$, and by $\mathcal{I}^{(S)}$ and $\rho(I^{(S)})$ accordingly.

Since these two structures act on disjoint column blocks of $\mathbf{X}$, a full ANCOVA structure is specified by a pair $I_{\mathcal{P}}^{(S)} = (I_{\mathcal{P}_A}, I^{(S)})$ with $I_{\mathcal{P}_A} \in \mathcal{I}_{\mathcal{P}_A}$ and $I^{(S)} \in \mathcal{I}^{(S)}$. For simplicity of notation, we write $I := I_{\mathcal{P}}^{(S)}$.

Majorant for clustering within U_A (one way ANOVA model ω_A)

Let $U_A = \{1, \dots, J\}$ be the index set of the J factor columns in $\mathbf{X}_A = [\mathbf{x}_{\alpha_1} \ \dots \ \mathbf{x}_{\alpha_J}]$. Each partition $\mathcal{P}_A = \{A_1, \dots, A_M\}$ of U_A , with $1 \leq M \leq J$, determines a clustered design matrix $\mathbf{X}_{I_{\mathcal{P}_A}}$ following equation (3.1.12):

$$\mathbf{X}_{I_{\mathcal{P}_A}} = [\mathbf{1}_n \ \mathbf{X}_A \mathbf{T}_A]$$

with $\mathbf{T}_A \in \{0, 1\}^{J \times M}$.

This matrix spans a linear subspace with its statistical dimensions according to equation (3.1.13):

$$\mathbb{L}_{I_{\mathcal{P}_A}} = \text{Col}(\mathbf{X}_{I_{\mathcal{P}_A}}), \quad d_{I_{\mathcal{P}_A}} = \text{rank}(\mathbf{X}_{I_{\mathcal{P}_A}}) \leq M.$$

The family of all these clustering structures $I_{\mathcal{P}_A}$ is

$$\mathcal{I}_{\mathcal{P}_A} = \{ I_{\mathcal{P}_A} : \mathcal{P}_A \text{ is a partition of } U_A \},$$

For a fixed number of clusters M , the number of distinct partitions of U_A is the *Stirling number of the second kind* $S(J, M)$ as in definition 1.4.2. Therefore, $\mathcal{I}_{\mathcal{P}_A}$ can be decomposed into disjoint subfamilies indexed by $M = 1, \dots, J$. The subfamily corresponding to M contains $S(J, M)$ structures, all of the same statistical dimension $d_{I_{\mathcal{P}_A}}$. Thus,

$$\ell(I_{\mathcal{P}_A}) = S(J, M)$$

Then, the majorant from equation (2.3.1) takes the explicit form:

$$\rho(I_{\mathcal{P}_A}) = M + \log(S(J, M)) \quad (3.2.1)$$

Finally, the size of the family is $|\mathcal{I}_{\mathcal{P}_A}| = B_J = \sum_{M=1}^J S(J, M)$.

Remark 3.2.2. *The Bell number $B_J = \sum_{M=1}^J S(J, M)$ grows rapidly with J , so the family $\mathcal{I}_{\mathcal{P}_A}$ already becomes very rich for moderate J . In practice, there are many competing clustering models to compare when minimizing the criterion (2.2.1), and searching over all of them quickly becomes computationally demanding.*

Majorant for unit sparsity of $U_1 \dots U_q$ (linear regression model ω_β)

Let $\{1, \dots, q\}$ be the index set of the q covariate columns in $\mathbf{X}_\beta = [\mathbf{x}_{\beta_1}, \dots, \mathbf{x}_{\beta_q}]$. Each active index set $S \subseteq \{1, \dots, q\}$ (or equivalently $S = \{j \in \{1, \dots, q\} : \delta(U_j) = 1\}$) with $|S| = s$ determines a sparse design matrix $\mathbf{X}_{I(S)}$ and therefore a linear subspace following equations (3.1.7) and (3.1.8):

$$\mathbf{X}_{I(S)} = [\mathbf{1}_n \quad \mathbf{X}_{\beta_{I(S)}}]$$

where $\mathbf{X}_{\beta_{I(S)}}$ keeps only the columns of $\mathbf{X}_\beta$ corresponding to units U_j with $j \in S$.

The associated linear subspace, with its statistical dimensions, is as follows:

$$\mathbb{L}_{I(S)} = \text{Col}(\mathbf{X}_{I(S)}), \quad d_{I(S)} = \text{rank}(\mathbf{X}_{I(S)}) \leq s + 1.$$

The family of all these unit-sparsity structures $I^{(S)}$ is:

$$\mathcal{I}^{(S)} = \{I^{(S)} : S \subseteq \{1, \dots, q\}\}$$

For a fixed sparsity level $s = |S|$, there are $\binom{q}{s}$ different choices of S . Therefore, the family $\mathcal{I}^{(S)}$ decomposes into subfamilies indexed by $s = 0, \dots, q$, where each contains all structures $I^{(S)}$ of the same statistical dimension $d_{I(S)}$. Thus,

$$\ell(I^{(S)}) = \binom{q}{s}$$

The majorant from equation (2.3.1) becomes:

$$\rho(I^{(S)}) = (s + 1) + \log \binom{q}{s} \quad (3.2.3)$$

Remark 3.2.4. *The case $s = 0$ means that there are no covariates in the model. Therefore, only the intercept remains, which corresponds to the null model ω_0 (1.3.10). In this case, the unit sparsity majorant reduces to the simplest value, that is $\rho(I^{(S)}) = 1$. This coincides with the clustering majorant $\rho(I_{\mathcal{P}_A}) = 1$ (3.2.1), for the extreme case $M = 1$.*

Finally, the size of the family is $|\mathcal{I}^{(S)}| = 2^q$.

Combined majorant (ANCOVA model $\omega_{A\beta}$)

Recall that a full ANCOVA structure is the pair

$$I = (I_{\mathcal{P}_A}, I^{(S)}), \quad I_{\mathcal{P}_A} \in \mathcal{I}_{\mathcal{P}_A}, \quad I^{(S)} \in \mathcal{I}^{(S)}$$

Each such pair determines the linear subspace:

$$\mathbb{L}_I = \text{Col}([\mathbf{1}_n \ (\mathbf{X}_A \mathbf{T}_A) \ X_{\beta_{I^{(S)}}}]),$$

and, using the dimension formula for the sum of subspaces [1, p. 156], its statistical dimension satisfies

$$d_I \leq M + s,$$

For fixed (M, s) , there are $S(J, M)$ admissible partitions of U_A into M blocks and $\binom{q}{s}$ admissible active sets of size s . Hence, by the multiplication principle [3, p. 28–29]

$$\ell(I) = S(J, M) \binom{q}{s}.$$

This gives the majorant:

$$\rho(I) = (M + s) + \log(S(J, M) \binom{q}{s}) = (M + s) + \log(S(J, M)) + \log \binom{q}{s} \quad (3.2.5)$$

Remark 3.2.6. *If $s = 0$, then no covariates are active. In this case, the ANCOVA structure reduces to the clustering structure of the one-way ANOVA part, and the majorant becomes $\rho(I_{\mathcal{P}_A}) = M + \log(S(J, M))$ (3.2.1).*

3.2.2. Two way ANOVA

Recall the full two way ANOVA model (1.3.7), with design matrix $\mathbf{X} = [\mathbf{1}_n \ \mathbf{X}_A \ \mathbf{X}_B \ \mathbf{X}_\gamma] \in \mathbb{R}^{n \times (1+J+K+JK)}$

Following equation (3.1.3), we partition $\mathbf{X}$ into disjoint units, as follows

$$\mathbf{X} = \left[\underbrace{\mathbf{x}_\mu}_{\text{intercept unit}}, \underbrace{\mathbf{x}_{\alpha_1}, \dots, \mathbf{x}_{\alpha_J}}_{U_A}, \underbrace{\mathbf{x}_{\beta_1}, \dots, \mathbf{x}_{\beta_K}}_{U_B}, \underbrace{\mathbf{x}_{\gamma_{11}}, \dots, \mathbf{x}_{\gamma_{JK}}}_{U_\gamma} \right].$$

This consists of three multi-column units: $U_A = \{1, \dots, J\}$, $U_B = \{1, \dots, K\}$ and $U_\gamma = \{1, \dots, JK\}$, associated with factors A, B and their interaction γ , respectively.

In this setting, we consider two independent structures: clustering within U_A , $I_{\mathcal{P}_A}$, and clustering within U_B , $I_{\mathcal{P}_B}$. Each structure gives rise to a separate family and thus to a corresponding majorant. We denote these by $\mathcal{I}_{\mathcal{P}_A}$ and $\rho(I_{\mathcal{P}_A})$ and by $\mathcal{I}_{\mathcal{P}_B}$ and $\rho(I_{\mathcal{P}_B})$ accordingly.

Majorant for clustering within U_A and U_B

Let $U_A = \{1, \dots, J\}$ be the index set of the J factor columns in $\mathbf{X}_A = [\mathbf{x}_{\alpha_1} \cdots \mathbf{x}_{\alpha_J}]$ and $U_B = \{1, \dots, K\}$ the corresponding index set of the K factor columns in $\mathbf{X}_B = [\mathbf{x}_{\beta_1} \cdots \mathbf{x}_{\beta_K}]$.

Each partition $\mathcal{P}_A = \{A_1, \dots, A_M\}$ of U_A , with $1 \leq M \leq J$, and $\mathcal{P}_B = \{B_1, \dots, B_N\}$ of U_B , with $1 \leq N \leq K$, defines a collapsed design matrix that spans a linear subspace following equations (3.1.12) and (3.1.13)

- In the case of no interactions (additive model ω_{AB}) that is :

$$\mathbf{X}_{I_{\mathcal{P}_{AB}}} = [\mathbf{1}_n \ \mathbf{X}_A \mathbf{T}_A \ \mathbf{X}_B \mathbf{T}_B], \text{ where } \mathbf{T}_A \in \{0, 1\}^{J \times M}, \mathbf{T}_B \in \{0, 1\}^{K \times N}.$$

$$\text{with } \mathbb{L}_{I_{\mathcal{P}_{AB}}} = \text{Col}(\mathbf{X}_{I_{\mathcal{P}_{AB}}}), \quad d_{I_{\mathcal{P}_{AB}}} = \text{rank}(\mathbf{X}_{I_{\mathcal{P}_{AB}}}) \leq M + N - 1,$$

- In the case of interactions (full model $\omega_{A \times B}$) that is:

$$\mathbf{X}_{I_{\mathcal{R}}} = [\mathbf{1}_n \ \mathbf{X}_A \mathbf{T}_A \ \mathbf{X}_B \mathbf{T}_B \ \mathbf{X}_{\gamma}(\mathbf{T}_A \otimes \mathbf{T}_B)], \text{ where } (\mathbf{T}_A \otimes \mathbf{T}_B) \in \{0, 1\}^{JK \times MN}$$

$$\text{with } \mathbb{L}_{I_{\mathcal{R}}} = \text{Col}(\mathbf{X}_{I_{\mathcal{R}}}), \quad d_{I_{\mathcal{R}}} = \text{rank}(\mathbf{X}_{I_{\mathcal{R}}}) \leq MN$$

Two families can arise from all candidate structures $I_{\mathcal{P}_A}$ and $I_{\mathcal{P}_B}$:

- the additive clustering family with all candidate paired structures $I_{\mathcal{P}_{AB}} = (I_{\mathcal{P}_A}, I_{\mathcal{P}_B})$, $I_{\mathcal{P}_A} \in \mathcal{I}_{\mathcal{P}_A}$, $I_{\mathcal{P}_B} \in \mathcal{I}_{\mathcal{P}_B}$

$$\mathcal{I}_{AB} = \{ I_{\mathcal{P}_{AB}} : \mathcal{P}_A \text{ is a partition of } U_A, \mathcal{P}_B \text{ is a partition of } U_B \}.$$

- the interaction clustering family with all admissible structures $I_{\mathcal{R}}$ coming from all possible Cartesian products of the two partitions:

$$\mathcal{I}_{\mathcal{R}} = \{ I_{\mathcal{R}} : \mathcal{R} = \mathcal{P}_A \times \mathcal{P}_B, \mathcal{P}_A \text{ is a partition of } U_A, \mathcal{P}_B \text{ is a partition of } U_B \},$$

In both cases, for fixed numbers of clusters (M, N) , the number of distinct partitions of U_A into M blocks is $S(J, M)$, and the number of partitions of U_B into N blocks is $S(K, N)$. Therefore, both $\mathcal{I}_{AB}$ and $\mathcal{I}_{\mathcal{R}}$ decompose into disjoint subfamilies indexed by (M, N) , where each one contains $S(J, M)S(K, N)$ structures of the same statistical dimension, $d_{I_{\mathcal{P}_{AB}}}$ and $d_{I_{\mathcal{R}}}$, respectively. In the additive case, a structure is a pair $(I_{\mathcal{P}_A}, I_{\mathcal{P}_B})$, so the two independent choices multiply by the multiplication principle. In the interaction case, each structure is indexed by the Cartesian product $\mathcal{R} = \mathcal{P}_A \times \mathcal{P}_B$, so the count equals its cardinality. Both cases give the same combinatorial term

$$\ell(I_{\mathcal{P}_{AB}}) = \ell(I_{\mathcal{R}}) = S(J, M) S(K, N).$$

This gives the interesting result that the two majorants; the one for the additive structures $\rho(I_{\mathcal{P}_{AB}})$, and the other for the interaction structures $\rho(I_{\mathcal{R}})$ differ only in the dimension part. Indeed, following equation (2.3.1), for the structure $I_{\mathcal{P}_{AB}}$ we obtain:

$$\begin{aligned}\rho(I_{\mathcal{P}_{AB}}) &= (M + N - 1) + \log[S(J, M) S(K, N)] \\ &= (M + N - 1) + \log[S(J, M)] + \log[S(K, N)]\end{aligned}\tag{3.2.7}$$

and accordingly for the structure $I_{\mathcal{R}}$

$$\begin{aligned}\rho(I_{\mathcal{R}}) &= MN + \log[S(J, M) S(K, N)] \\ &= MN + \log S(J, M) + \log S(K, N)\end{aligned}\tag{3.2.8}$$

Finally, the size of both families is $|\mathcal{I}_{AB}| = |\mathcal{I}_{\mathcal{R}}| = B_J \times B_K$

The majorants (3.2.7) and (3.2.8) highlight a clear difference between additive $I_{\mathcal{P}_{AB}}$ and interaction structures $I_{\mathcal{R}}$. In both cases, the combinatorial richness of the two clustering families $\mathcal{I}_{AB}$ and $\mathcal{I}_{\mathcal{R}}$ is the same. However, the statistical dimension changes from $M + N - 1$ in the additive structure to MN in the interaction structure. As a consequence, the interaction structures receive a larger complexity penalty than their additive ones with the same (M, N) .

In practice, the criterion $C_{\kappa}(I)$ is expected to favor $I_{\mathcal{P}_{AB}}$ unless $I_{\mathcal{R}}$ provides a sufficiently large improvement in goodness of fit. Specifically, only if it reduces the RSS by more than $\kappa\sigma^2(M - 1)(N - 1)$. A short calculation can show that.

$$\begin{aligned}C_{\kappa}(I_{\mathcal{R}}) - C_{\kappa}(I_{\mathcal{P}_{AB}}) &= (\text{RSS}_{I_{\mathcal{R}}} - \text{RSS}_{I_{\mathcal{P}_{AB}}}) + \kappa\sigma^2(MN - (M + N - 1)) \\ &= (\text{RSS}_{I_{\mathcal{R}}} - \text{RSS}_{I_{\mathcal{P}_{AB}}}) + \kappa\sigma^2(M - 1)(N - 1).\end{aligned}\tag{3.2.9}$$

The goodness of fit improvement is $G = \text{RSS}_{I_{\mathcal{P}_{AB}}} - \text{RSS}_{I_{\mathcal{R}}} > 0$ then from equation (3.2.9) the interaction structure is chosen if

$$-G + \kappa\sigma^2(M - 1)(N - 1) < 0 \Leftrightarrow G > \kappa\sigma^2(M - 1)(N - 1)$$

In other words, the additional complexity of the interaction is accepted only if the reduction in RSS is worth its penalty cost.

3.3. Motivation for adjusting the majorant

Recall that the majorant $\rho(I)$ (2.3.1) was chosen large enough to control a very rich family $\mathcal{I}$ of structures and to penalize not only how large the selected structure is, but also how large the search space around that structure is. With $\ell(I)$ structures all at the same dimension d_I , each must be penalized by $\log \ell(I)$ to keep $\sum_{I \in \mathcal{I}} e^{-\nu\rho(I)}$ bounded. However, in the one-way and two-way ANOVA settings of this thesis, the number of factor levels is fixed. Any intermediate clustering structure is combinatorially larger than

the unique maximum clustering structure, that is, the full model. The full model, being alone in its own subfamily, is "cheaper" relative to any intermediate clustering. The combinatorial term $\log \ell(I)$ therefore overpenalises any intermediate clustering structure in these settings, and the following comparison makes this precise.

Consider the one way ANOVA model with a factor of J levels. Let I_{clust} be an intermediate clustering structure with $1 < M < J$ blocks and I_{full} the full model. We cannot find a tuning parameter $\kappa > 0$ such that

$$C_\kappa(I_{clust}) < C_\kappa(I_{full}) \quad (3.3.1)$$

Notice that $RSS_{I_{clust}} \geq RSS_{I_{full}}$ always holds, as I_{clust} is nested in I_{full} .

Since $\rho(I_{full}) = J + \log S(J, J) = J$, equation (3.3.1) becomes:

$$(RSS_{I_{clust}} - RSS_{I_{full}}) + \kappa \sigma^2(M - J + \log S(J, M)) < 0$$

Since $RSS_{I_{clust}} - RSS_{I_{full}} \geq 0$ for the criterion to choose I_{clust} the second term must be negative

$$\kappa \sigma^2(M - J + \log S(J, M)) < 0$$

As $\kappa > 0$, it does not affect the sign, so the recovery of I_{clust} cannot be fixed by only tuning κ ; it also requires

$$M + \log S(J, M) < J. \quad (3.3.2)$$

By definition 1.4.1 it always follows that $1 \leq M \leq J$. Thus, the combinatorial term $\log S(J, M)$ controls recovery. Specifically, it pushes the penalty above the dimension gap $J - M$ and determines the sign of equation (3.3.1)

A similar result holds for the two way ANOVA model. In both the additive and the full case, the clustered and reference structures share the same combinatorial term and differ only in dimensions, so the same argument applies unchanged. We omit the details.

A way to address this is to add a weight $\lambda \in [0, 1]$ to the combinatorial part $\log \ell(I)$. Recall that condition (C2) $\rho(I) \gtrsim d_I$, requires the penalty to be at least of the order of the dimension; weighting only the combinatorial term keeps this, since the dimension part is left untouched. This gives the adjusted majorant:

$$\rho_\lambda(I) = d_I + \lambda \log \ell(I), \quad \lambda \in [0, 1] \quad (3.3.3)$$

The penalized criterion $C_\kappa(I)$ (2.2.1) then becomes:

$$C_{\kappa, \lambda}(I) = RSS_I + \kappa \sigma^2 \rho_\lambda(I) = RSS_I + \kappa \sigma^2 [d_I + \lambda \log \ell(I)] \quad (3.3.4)$$

The criterion now has two tuning parameters: the overall penalty strength $\kappa > 0$ and the complexity weight $\lambda \in [0, 1]$, which controls how strongly the combinatorial term contributes. The weight λ lies between two limiting cases.

- At $\lambda = 1$ we recover the full theoretical majorant $\rho_1(I) = d_I + \log \ell(I)$.
- At $\lambda = 0$ the combinatorial term vanishes and we obtain a simplified majorant, dimension-based $\rho_0(I) = d_I$, which retains only the statistical dimension of $\mathbb{L}_I$ and treats all structures equally at the same dimension.
- Intermediate values $0 < \lambda < 1$ retain the combinatorial term, but with a reduced weight.

Neither extreme can be assumed optimal in advance; both κ and λ must be tuned, as the two parameters play different roles.

Remark 3.3.5. *This adjustment should be understood as a practical finite sample calibration, rather than a new theoretical recovery result. The effect of κ and λ is studied empirically in the simulations chapter.*

3.3.1. Connection to classical criteria.

The selected value $\lambda = 0$ is worth a closer look, because it gives the criterion a familiar form. At $\lambda = 0$ the adjusted majorant reduces to $\rho_0(I) = d_I$, so the criterion becomes

$$C_{(\kappa,0)}(I) = \text{RSS}_I + \kappa d_I,$$

which penalizes a structure only through its dimension.

This is in the same spirit as the classical model selection criteria for low-dimensional Gaussian linear models, such as AIC [20] and Mallows' C_p [21].

AIC	Mallows' C_p
$n \ln \left(\frac{\text{RSS}_p}{n} \right) + 2p$ For Gaussian linear models the log-term simplifies and $\text{AIC} = C_p + n$	$\frac{\text{RSS}_p}{\sigma^2} + 2p - n$
If σ^2 is unknown, it is estimated locally: $\hat{\sigma}^2 = \frac{\text{RSS}_p}{n}$	If σ^2 is unknown, it is estimated from the full model: $\hat{\sigma}^2 = \frac{\text{RSS}_{\text{full}}}{n - p}$

Table 3.4.: AIC and Mallows' C_p formulas for Gaussian linear models. Here, RSS_p is the residual sum of squares of the fitted candidate model and p is the number of free parameters.

Dividing $C_{(\kappa,0)}(I)$ by σ^2 gives

$$\frac{\text{RSS}_I}{\sigma^2} + \kappa d_I$$

.

- When σ^2 is known, this is structurally identical to AIC, with d_I playing the role of the number of free parameters p .
- When σ^2 is unknown, we use the unbiased estimator from the full model $\hat{\sigma}^2 = RSS_{full} / (n - p)$. This is structurally identical to Mallows's C_p up to the additive constant n .

4. Simulation study in ANCOVA models

In this chapter, we conduct a simulation study in R to empirically study the performance of the criterion $C_{\kappa,\lambda}(I)$ (3.3.4). We start with synthetic data generated from a simple one way ANOVA model, for which the true clustering structure is known by construction. We then extend the study to ANCOVA by adding covariates q , whose true sparsity is also known by construction.

4.1. Simulation setup for the one way ANOVA model

We consider the full one way ANOVA model ω_A (1.3.5) where factor A has $J = 5$ levels and a balanced design with $n_j = 20$ observations per level, giving a total sample size of $n = 100$. We generate $N_{sim} = 200$ independent observations according to the following model:

$$Y_{ij} = \mu + \alpha_i + \sigma e_{ij}, \quad e_{ij} \sim \mathcal{N}(0, 1),$$

for $i = 1, \dots, 5$ and $j = 1, \dots, 20$, where $\mu = 0$ and $\sigma = 1$.¹

The true structure $I_{\mathcal{P}_{true}}$ is given by the following partition of the levels of factor A:

$$\mathcal{P}_{true} = \{\{1, 3\}, \{2, 5\}, \{4\}\}.$$

We take the true cluster means to be

$$\alpha_1 = \alpha_3 = 0, \quad \alpha_2 = \alpha_5 = 1.5, \quad \alpha_4 = 3.$$

The signal-to-noise ratio, also called the standardized mean difference, is $\delta/\sigma = 1.5$. In the sense of effect sizes, $\delta/\sigma = 1.5$ corresponds to a large separation relative to the noise level [8, 9]. For recovering the structure $I_{\mathcal{P}_{true}}$, what matters is the same ratio reinforced by the group size n_j . Each level mean is estimated from $n_j = 20$ observations, so its standard error is $\sigma/\sqrt{n_j}$ and the separation on the level mean scale becomes $(\delta/\sigma)\sqrt{n_j}$. Here, this equals $1.5\sqrt{20} \approx 6.7$, that is, the cluster means are about 6.7 standard errors apart, so they are statistically distinguishable. Therefore, the simulated signal is strong

¹This is the scalar version of the one way ANOVA model of (1.3.5).

enough for a well-performing method to detect it [17].

The goal is to minimize $C_{\kappa,\lambda}(I)$ (3.3.4) for this one way ANOVA model over the full family of all admissible structures, whose cardinality is $|\mathcal{I}| = B_5 = 52$. Using equation 3.2.1 we get the following:

$$C_{\kappa,\lambda}(I) = \text{RSS}_I + \kappa\sigma^2[M + \lambda \log S(J, M)] \quad (4.1.1)$$

4.1.1. Tuning via recovery rates

We start with tuning (κ, λ) using recovery rates of the known true structure $I_{\mathcal{P}_{true}}$. For each (κ, λ) on a predetermined grid, we count how often, in $N_{\text{sim}} = 200$ runs, the criterion selects $I_{\mathcal{P}_{true}}$, and we report this recovery rate. Since $\kappa > 0$, it is searched on a logarithmic grid running from $\kappa = 0.1$ to $\kappa = 20$, which covers small to large penalties evenly. In contrast, λ is searched on a linear grid within its interval $[0, 1]$.

A representative part of these results is reported in table 4.1. The full table can be found in the appendix A.1.

κ	λ				
	0	0.2	0.4	0.6	0.8
0.585	0.310	0.265	0.190	0.000	0.000
1.414	0.575	0.500	0.380	0.040	0.000
2.199	0.755	0.670	0.520	0.075	0.000
3.420	0.875	0.830	0.705	0.115	0.000
5.318	0.950	0.910	0.840	0.185	0.000
8.270	0.980	0.965	0.900	0.260	0.000
12.861	0.935	0.905	0.885	0.295	0.000
20	0.815	0.735	0.655	0.265	0.000

Table 4.1.: Recovery rates of the $I_{\mathcal{P}_{true}}$ among the 52 admissible candidate structures, over $N_{\text{sim}} = 200$ simulations, for a range of values of (κ, λ) .

We observe the following.

- Recovery rates increase with κ up to the moderate value $\kappa = 8.270$. For larger κ they decrease, since a stronger penalty can cause under-clustering. For every merge of two true clusters, the RSS increases by about $(1.5)^2 \times 20 = 45$ squared error units, while saving $\kappa \times 20 = 20$ in the largest penalty case of this grid. Because the RSS varies per trial, in cases where this increase becomes less than 20, under-clustering occurs, and from the results we can see that this happens in $\approx 18.5\%$ of the runs.

- A different behavior is shown for λ . Here, recovery rates increase as λ decreases, reaching their highest value at $\lambda = 0$. A closer look at the combinatorial cost explains this. From table 4.2 we can see that this cost is the smallest in the extreme cases $M = 1$ and $M = 5$, and becomes larger in the intermediate clustering scenarios $1 < M < 5$, reaching its highest value at $M = 3$, where $I_{\mathcal{P}_{true}}$ belongs. Therefore, λ plays a crucial role in how expensive the combinatorial term becomes, and for this simulation scenario sets it to zero.

M	$S(5, M)$	$\log S(5, M)$
1	1	0
2	15	≈ 2.7081
3	25	$\approx$ 3.2189
4	10	≈ 2.30016
5	1	0

Table 4.2.: Stirling's number of the second kind for $J = 5$ and $M = 1, 2, 3, 4, 5$ and its log values

4.1.2. Cross-validation tuning

The above tuning relies on knowing the true structure since we counted how often the criterion was recovered $I_{\mathcal{P}_{true}}$, which is only possible in a simulation where the truth is known by construction. In practice, with real data, the true structure is unknown, so recovery cannot be measured and some other rule is needed to choose (κ, λ) . Cross-validation provides such a rule [10]. Instead of asking how often the criterion recovers the truth, it asks how well the selected structure predicts data that were not used to fit it. The benefit is that this can be computed from the data alone, without knowledge of $I_{\mathcal{P}_{true}}$, so the same procedure applies equally to the simulation and to the real-data study of the next chapter.

We use k -fold cross-validation. The n observations are split into k groups of equal size, called folds, denoted $F_1, \dots, F_k$. For a given pair (κ, λ) , we treat each fold F_j in turn as the validation set, while the remaining $k - 1$ folds form the training set. For each fold F_j , we proceed as follows:

- **Train:** using only the training set (all observations not in F_j), select the structure $\hat{I}_{-j}$ that minimizes the criterion $C_{\kappa, \lambda}(I)$, over the full family $|\mathcal{I}|$, and fit it to obtain the predictions $\hat{y}_i^{(-j)}$ for each observation i from the structure $\hat{I}_{-j}$.
- **Validate:** for each observation in the validation set $i \in F_j$, compute the squared prediction error $(y_i - \hat{y}_i^{(-j)})^2$.

For each fold, we first average the squared prediction errors over the points in that fold

and then average these fold errors over the k folds. This gives the cross validation error:

$$\text{CV}(\kappa, \lambda) = \frac{1}{k} \sum_{j=1}^k \frac{1}{|F_j|} \sum_{i \in F_j} (y_i - \hat{y}_i^{(-j)})^2, \quad (4.1.2)$$

where $|F_j|$ is the number of observations in fold F_j . The chosen tuning is the pair that minimizes this error,

$$(\hat{\kappa}, \hat{\lambda}) = \arg \min_{(\kappa, \lambda)} \text{CV}(\kappa, \lambda).$$

For each of the independently generated data sets and for each (κ, λ) on the same grid as above, we apply this k -fold cross validation for $k=5$. Since the complete sample has $n = 100$ observations, each fold contains $|F_j| = 20$ observations, and the corresponding training set has the remaining 80 observations. Then, the following cross validation error from equation (4.1.2) is calculated:

$$\text{CV}(\kappa, \lambda) = \frac{1}{5} \sum_{j=1}^5 \frac{1}{20} \sum_{i \in F_j} (y_i - \hat{y}_i^{(-j)})^2.$$

We then average the above errors over the $N_{\text{sim}} = 200$ simulations and choose the tuning pair $(\hat{\kappa}, \hat{\lambda})$ with the smallest average one.

$$(\hat{\kappa}, \hat{\lambda}) = \arg \min_{(\kappa, \lambda)} \overline{\text{CV}(\kappa, \lambda)}. \quad (4.1.3)$$

The procedure selects

$$\hat{\kappa} = 5.318, \quad \hat{\lambda} = 0, \quad \text{with } \text{CV}(\hat{\kappa}, \hat{\lambda}) = 1.059.$$

Figure 4.1 shows the cross-validation error $\text{CV}(\kappa, \lambda)$ as a function of $\log \kappa$, with a separate colored curve for each value of λ . The left panel shows the full range, and the right panel zooms in around the minimum, making it clear that the error attains a well-defined minimum at the chosen $(\hat{\kappa}, \hat{\lambda})$. The full table of the averaged cross-validation values for all pairs of (κ, λ) is reported in appendix A.2.

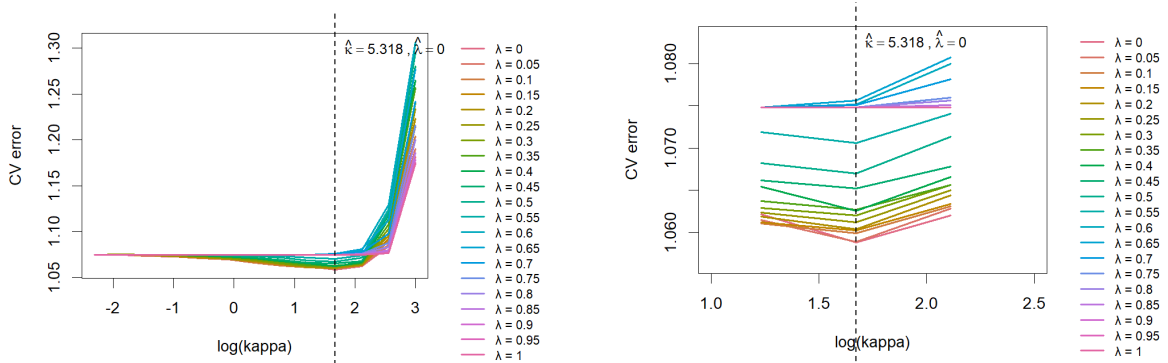


Figure 4.1.: Averaged cross-validation error $\text{CV}(\kappa, \lambda)$ over $N_{\text{sim}} = 200$ simulated datasets with $k = 5$ folds against $\log \kappa$, with one colored curve per value of λ . Left: full range. Right: zoom near the minimum.

We observe that the value $\hat{\kappa} = 5.318$ is somewhat lower than the previous $\kappa = 8.27$ at which the recovery first reaches 100%. This makes sense since the two criteria measure different things. Recovery counts how often the selected structure is exactly $I_{\mathcal{P}_{\text{true}}}$, and this count keeps increasing with κ until the method almost always avoids overfitting, so it prefers a stronger penalty. Cross-validation instead measures how well the selected structure predicts new observations. Once the penalty is large enough to recover the true structure, the prediction error is essentially at its lowest, and a further increase κ does not improve it. The two therefore do not point to an identical value, but agree on the important features: a moderate κ and $\lambda = 0$.

4.1.3. Selection using both tuning values

Finally, for each $N_{\text{sim}} = 200$ simulated data set, we minimize the criterion $C_{\kappa,\lambda}(I)$ on all $B_5 = 52$ candidate structures, using both the cross-validation pair $(\hat{\kappa}, \hat{\lambda}) = (5.318, 0)$ and recovery-optimal tuning $(\kappa, \lambda) = (8.270, 0)$. In table 4.3 we record, for each tuning, the proportion of simulations in which each structural class (null, 2-cluster, true 3-cluster, wrong 3-cluster, 4-cluster, full) is selected as the minimizer of $C_{\kappa,\lambda}(I)$.

Tuning	κ	λ	Null	$M = 2$	True	Wrong 3	$M = 4$	Full
Cross-validation	5.318	0	0	0	0.950	0.005	0.045	0
Recovery-optimal	8.270	0	0	0.015	0.980	0	0.005	0

Table 4.3.: Proportion of times each structural class minimizes $C_{\kappa,\lambda}(I)$ over all $B_5 = 52$ candidate structures, across $N_{\text{sim}} = 200$ simulations. The first row uses the cross-validation tuning of section 4.1.2; the second uses the recovery-optimal tuning of section 4.1.1

We observe the following.

- The selection rate of $I_{\mathcal{P}_{\text{true}}}$ is 95% in cross-validation tuning and 98% in recovery optimal. Therefore, the tuning chosen without the truth performs only slightly below the tuning that uses it. In both cases, the criterion recovers the true structure in the vast majority of datasets.
- The choice of a 4-cluster structure is an over-clustering mistake that is more likely at a smaller value of κ . It occurs in 4.5% of the simulations under cross-validation tuning and only in 0.5% with the recovery-optimal one.
- Similarly, the choice of a 2-cluster structure is an under-clustering mistake that appears only at a larger κ , so the 1.5% occurs only at the recovery-optimal tuning.
- The wrong 3-cluster structure, at 0.5%, is not related to the choice of κ , as it has the same statistical dimensions as the true one. In one of the 200 simulations, the empirical level means accidentally made the wrong partition fit slightly better than the true one.

4.2. The role of $M_{\text{true}} = 2$ and large J for the theoretical majorant

Finally, recall the inequality $M + \log S(J, M) < J$ from equation (3.3.2) in section 3.3. Interestingly, for fixed M and $J \rightarrow \infty$, the Stirling number of the second kind satisfies

$$S(J, M) \sim M^J / M!$$

Using this, for large J equation (3.3.2) reads as follows

$$J \log M - \log(M!) < J - M \implies \log M < 1 - \frac{M - \log(M!)}{J} \xrightarrow{J \rightarrow \infty} \log M < 1,$$

which holds if and only if $M < e \approx 2.718$. Since $M \in \mathbb{Z}^+$, only $M_{\text{true}} = 2$ satisfies this condition. Therefore, we suspect that for some $\kappa > 0$ the theoretical majorant $\rho_1(I) = M + \log S(J, M)$ can recover a true 2-cluster structure for large J .

An extra simulation from the same one way ANOVA setting, from section 4.1 with varying both the number of factor levels $J \in \{5, 10, 20, 40\}$ and the true number of clusters $M_{\text{true}} \in \{2, \lfloor J/4 \rfloor, \lfloor J/2 \rfloor, \lfloor 3J/4 \rfloor, J - 1\}$ shows this interesting result.

Remark 4.2.1. *We choose these representative values for M_{true} , to cover simple, moderate, complex and close to full clustering scenarios.*

We fix $\lambda = 1$ and moderate $\kappa = 5.32$. We choose this κ value only as a fixed reference, to keep the comparison between different J and M_{true} values on a common footing, and we do not claim that it is optimal in this setting.

Since the total number of partitions grows with J through the Bell number B_J , an exhaustive search becomes computationally infeasible for large J . Instead, for each pair (J, M_{true}) we minimize

$$C_{\kappa,1}(I) = RSS_I + \kappa \sigma^2 [M + \log S(J, M)]$$

over four representative candidate structures: the full, the null, the true M -cluster, and a wrong M -cluster model.² The wrong model has the same number of clusters as the true one, so it carries an identical dimension and an identical combinatorial penalty. Table 4.4 shows a representative part of the recovery rates for different (J, M_{true}) together with the theoretical majorant values $\rho_{\text{true}} = M_{\text{true}} + \log S(J, M_{\text{true}})$ and $\rho_{\text{full}} = J$. For the complete table, refer to A.3.

²This is obtained by a cyclic shift of the true cluster labels and when the cyclic shift collapses the labels into a different number of clusters, the corresponding (J, M_{true}) combination is skipped.

J	M_{true}	ρ_{true}	ρ_{full}	Recovery
5	2	4.71	5	0.345
5	3	6.22	5	0.000
10	2	8.24	10	0.690
10	5	15.66	10	0.000
20	2	15.17	20	0.900
20	10	39.41	20	0.000
40	2	29.03	40	0.980
40	20	94.17	40	0.000
60	2	42.90	60	0.995
60	30	154.29	60	0.000
80	2	56.76	80	1.000
80	40	217.88	80	0.000
90	2	63.69	90	1.000
90	45	250.67	90	0.000

Table 4.4.: Values of the theoretical majorant for the true cluster M_{true} and the full model, and recovery rates, over $N_{\text{sim}} = 200$ runs, varying both J and M_{true} at fixed $\kappa = 5.32$.

Indeed, we can see that for $M_{\text{true}} = 2$ the theoretical majorant achieves positive recovery, increasing from 34.5% at $J = 5$ to 100% at $J = 80$. Figure 4.2 visualizes this monotonic increase of the recovery rate for $M_{\text{true}} = 2$ with different values of J .

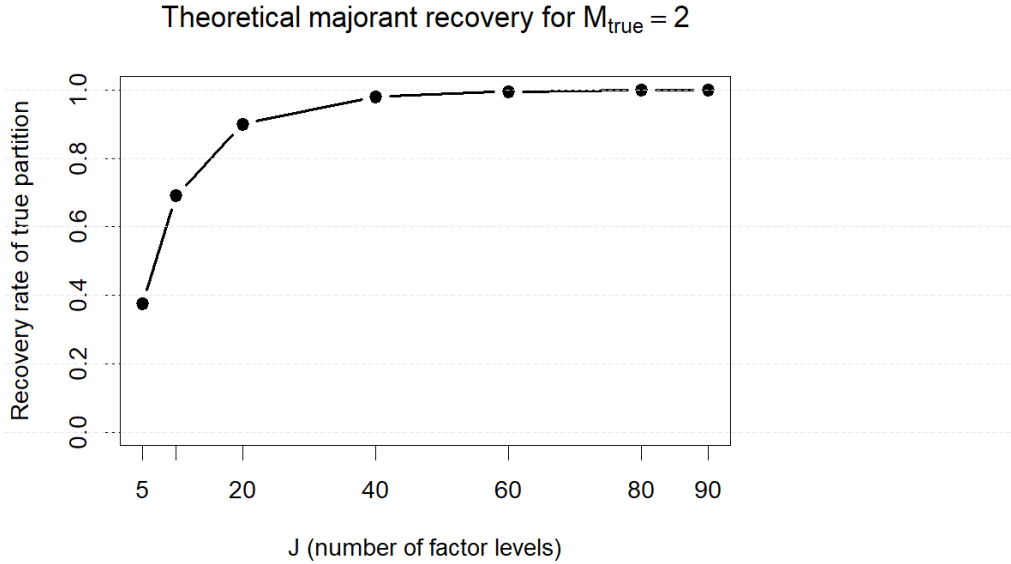


Figure 4.2.: Recovery rate of the criterion using the theoretical majorant over $N_{\text{sim}} = 200$ runs, as a function of J , for $M_{\text{true}} = 2$, at fixed $\kappa = 5.32$

We highlight that these results are based on a moderate κ . We expect that for very large κ the recovery rates go down, since a stronger penalty can cause under-clustering, as we saw in section 4.1.1. Similarly, for very small κ they go down as well, since the penalty is then too weak to control the fit and over-clustering occurs.

4.3. Extending the study to the ANCOVA model

4.3.1. Simulation setup

We now extend the one way ANOVA setting of section 4.1 by adding $q = 11$ covariates, with no interactions present. The observations are generated according to the ANCOVA model $\omega_{A\beta}$ from equation (1.3.1):

$$Y_{ij} = \mu + \alpha_i + \sum_{k=1}^{11} \beta_k x_{ijk} + \sigma e_{ij}, \quad e_{ij} \sim \mathcal{N}(0, 1),$$

where the covariate values x_{ij} are drawn independently from $\mathcal{N}(0, 1)$.

The true factor structure is the same $I_{\mathcal{P}_{true}}$ as in section 4.1 and the true covariate structure $I^{(S_{true})}$ is imposed by $S_{true} = \{1, 2, 3, 4, 5\}$, which means that of the covariates $q = 11$, only the first five are active. The true ANCOVA structure is therefore the pair $I_{true} = (I_{\mathcal{P}_{true}}, I^{(S_{true})})$

We take the true covariates to be well separated from zero:

$$\beta_1 = 2, \beta_2 = -2, \beta_3 = 1.5, \beta_4 = -1.5, \beta_5 = 2.5.$$

Each true coefficient is estimated with standard error $se(\hat{\beta}_k) = \sigma / \|\tilde{\mathbf{x}}_k\|$, where $\tilde{\mathbf{x}}_k = \mathbf{x}_k - \mathbf{P}_{Col(\mathbf{X}_A)} \mathbf{x}_k$. Since the covariates are drawn independently from $\mathcal{N}(0, 1)$, we have $\|\tilde{\mathbf{x}}_k\| = \sqrt{n}$, so $se(\hat{\beta}_k) \approx \sigma / \sqrt{n}$. The smallest active signal is $|\beta_{min}| = 1.5$, and hence, on the standardized scale $|\beta_{min}| / se(\hat{\beta}_k) \approx (|\beta_{min}| / \sigma) \sqrt{n} = 1.5 \sqrt{100} \approx 15$, that is, the weakest covariate signal is about 15 standard errors from zero.

The goal is to minimize $C_{\kappa, \lambda}(I)$ (3.3.4) for this ANCOVA setting on the candidate family $\mathcal{I}$, which now consists of all the paired structures $I = (I_{\mathcal{P}}, I^{(S)})$, and has size $B_5 \times 2^q = 52 \times 2^{11} \approx 1.06 \times 10^5$. Using equation (3.2.5) we get the following:

$$C_{\kappa, \lambda}(I) = \text{RSS}_I + \kappa \left[(M + s) + \lambda \left(\log S(J, M) + \log \binom{q}{s} \right) \right], \quad (4.3.1)$$

4.3.2. Tuning via recovery rates

For each pair (κ, λ) on the same grid as in section 4.1.1, we count over $N_{\text{sim}} = 200$ simulated datasets how often the criterion recovers the true ANCOVA structure, and we

report this recovery rate.

We do not tune by cross-validation here. Our aim is different: we want to measure how often the true structure is recovered and, in particular, to observe how λ acts on the two combinatorial terms, one for the clustering and one for the covariates. Moreover, the search here is much larger, so cross-validation would add an extra loop, which would make the exhaustive procedure considerably more demanding.

We first report the recovery rates of $I_{\mathcal{P}_{true}}$ and $I^{(S_{true})}$ using heat maps, placed side by side, so that the behavior of (κ, λ) can be compared between the two structures. Figure 4.3 shows the most important values; full heat maps can be found in A.4

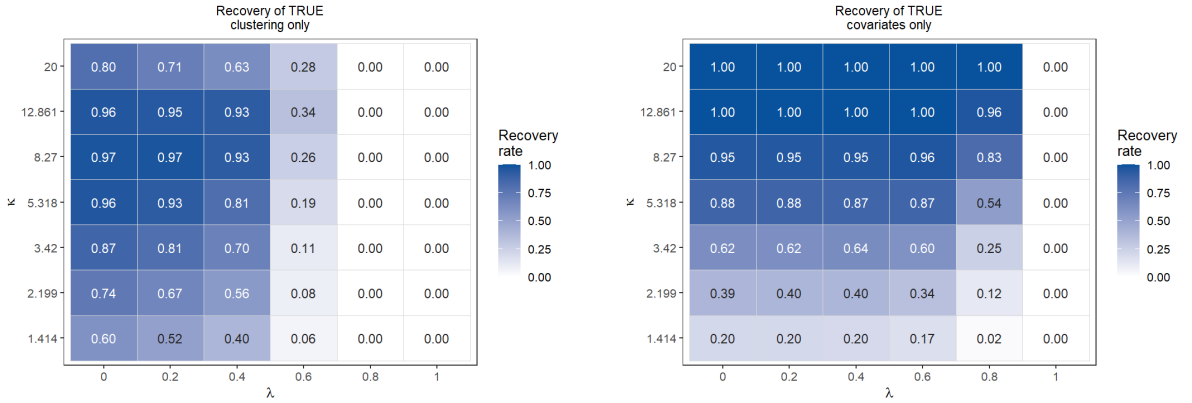


Figure 4.3.: Recovery rates of the $I_{\mathcal{P}_{true}}$ on the left and of the $I^{(S_{true})}$ on the right among all admissible candidate structures, over $N_{sim} = 200$ simulations, for a selected range of values of (κ, λ) .

For the cluster structure, the behavior is essentially similar as in section 4.1.1: recovery increases with κ up to the moderate value $\kappa = 8.27$ and is highest at $\lambda = 0$. We therefore do not repeat the discussion and instead focus on the sparsity structure.

For the sparsity structure, we observe the following:

- Recovery keeps increasing up to the largest $\kappa = 20$. A strong penalty normally would cause underfitting, but this does not happen here because the covariates have a really strong signal. Dropping an active covariate increases RSS by $(1.5)^2 \times 100 = 225$ square units to save $\kappa \times 20 = 20$ in penalty. This is a poor trade; therefore, the criterion will not do that.
- There is a different behavior of λ compared to the clustering case. Recovery stays high across a large part of the λ grid, up to about $\lambda \approx 0.8$. This is an effect of the combinatorial cost. From table 4.5 we can see that this cost is the smallest in the extreme cases $s = 0$ and $s = 11$ and grows with each added covariate.

The maximum value is at $s = 5$, where $I^{(S_{true})}$ lies. This explains why recovery collapses to 0% for $\lambda = 1$.

s	$\binom{11}{s}$	$\log \binom{11}{s}$
0 and 11	1	0
1 and 10	11	≈ 2.3979
2 and 9	55	≈ 4.0073
3 and 8	165	≈ 5.1059
4 and 7	330	≈ 5.7991
5 and 6	462	≈ 6.1356

Table 4.5.: Binomial coefficients $\binom{11}{s}$ for $s = 0, \dots, 11$ and their logarithmic values.

- The most interesting part here is that the combinatorial cost cannot distinguish the true structure from its near competitors: the addition of one noise covariate ($s = 6$) has the same cost as the true one ($s = 5$). Therefore, λ has only a mild effect on the selection of covariates, and it is κ that dominates the decision.

We finally report the recovery rates for the full true ANCOVA structure $I_{true} = (I_{\mathcal{P}_{true}}, I^{(S_{true})})$, counting a success only when both parts of this ordered pair are recovered simultaneously. See figure 4.4

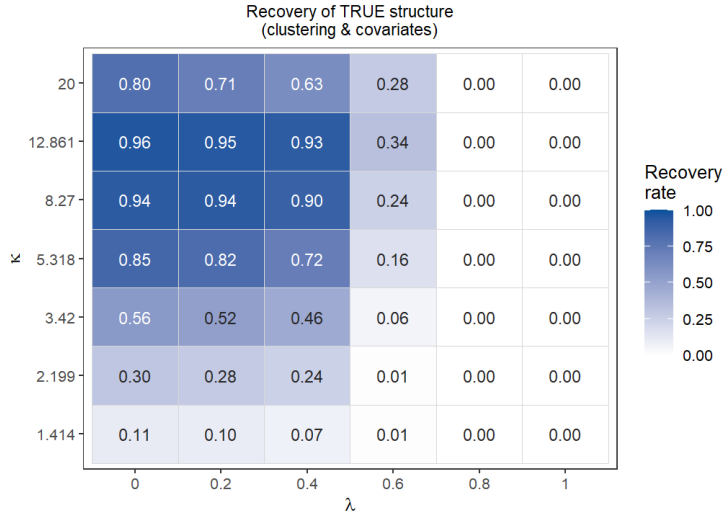


Figure 4.4.: Recovery rates of the $I_{\mathcal{P}_{true}}$ and $I^{(S_{true})}$ simultaneously, among all admissible candidate structures, over $N_{\text{sim}} = 200$ simulations, for a selected range of values of (κ, λ) .

The combined recovery rates make clear that the two structures need different tunings. The sparsity structure is recovered best towards large κ values while clustering needs a

moderate κ ; and while λ must stay close to 0 for the clustering, it has a little influence on the sparsity. The joint optimum is therefore a compromise dictated by the most fragile structure of the two, which here is the clustering. Therefore, the resulting sweet spot is $(\kappa, \lambda) = (12.861, 0)$.

5. Real data study in two way ANOVA models

5.1. The Palmer penguins dataset

The Palmer penguins dataset contains real-world body size measurements for three penguin species breeding on islands of the Palmer Archipelago in Antarctica. These data were collected between 2007 and 2009 by Dr. Kristen Gorman in collaboration with the Palmer Station Long-Term Ecological Research Network (LTER) [26, 27]. It is a realistic data set, large enough with $n = 344$ observations across 8 variables, it contains unequal sample sizes across groups and includes a small number of missing values. Additionally, it includes three factor variables (species, island, sex), offering nice opportunities for exploring the considered ANOVA structures of this thesis. Table 5.1 gives an overview of the eight variables included in this dataset. [22]

Variable	Type	Description
species	factor	Penguin species: Adélie, Chinstrap, or Gentoo
island	factor	Island: Biscoe, Dream, or Torgersen
bill_length_mm	numeric	Length of the dorsal ridge of the bill (mm)
bill_depth_mm	numeric	Depth of the bill (mm)
flipper_length_mm	integer	Length of the flipper (mm)
body_mass_g	integer	Body mass (g)
sex	factor	Sex of the penguin: female or male
year	integer	Year of study: 2007, 2008, or 2009

Table 5.1.: Overview of the 8 variables in the Palmer penguins dataset.

A first visual exploration of the dataset through the provided principal component analysis (PCA) bi-plot in [22, Figure 5A], already suggests that Gentoo penguins have a higher body mass than Adélie and Chinstrap, while the latter two may be more similar to each other. This visual impression is further explored, through a classical two way ANOVA analysis [25], when body mass is taken as the response variable with sex and species as factors, Adélie and Chinstrap penguins turn out to be statistically indistinguishable from one another, while both differ clearly from the Gentoo. Additionally, interactions of the two factor effects are present. This makes the Palmer penguins a nice real data set for applying the framework developed in this thesis for a two way ANOVA

model where clustering and interactions are present.

5.1.1. Data preprocessing

The penguins dataset originally contains 344 observations, of which 11 have missing values for the variables sex and body mass, thus we remove all incomplete cases, leaving $n = 333$ observations for the analysis. Additionally, the response variable body mass is recorded in grams, the classical ANOVA analysis keeps it like that. However, for our projection framework application we decide to rescale the response variable to kilograms. This is a linear transformation and does not change any of the structural conclusion of the model. It leaves the projection subspaces $\mathbb{L}_I$ and the selected structure $\hat{I}$ unchanged and it brings the RSS from order of 10^6 down to order 1, which is numerically more convenient when evaluating criterion $C(I)$. Additionally, the projected means and the corresponding cluster separations are expressed on a scale that is easier to interpret in practice (for example, differences of 0.5–1 kg between groups instead of 500–1000 g).

5.2. Model set up

5.2.1. Preliminary classical analysis

We first briefly mention here the main results on the classical two way ANOVA analysis of [25] applied to the Palmer penguins dataset with body mass (in grams) as the response variable and species and sex as the two factors. First of all, the standard assumptions of normality of residuals and homoscedasticity are verified and only two mild outliers are detected in the Chinstrap group via box-plots, but are judged not extreme enough to bias the results. The full two way ANOVA model (with interactions) is then fitted. All three effects are highly significant at the 5% % level: both main effects are present, body mass differs significantly between sexes and between species, and the significant interaction term ($p = 0.000197$) indicates that the effect of species on body mass is not the same for females and males. The full model with interaction is therefore retained. Finally, a post-hoc Tukey HSD test on the species factor reveals that body mass is not significantly different between Adélie and Chinstrap penguins (adjusted $p = 0.824$), whereas Gentoo penguins differ significantly from both other species (adjusted $p < 0.001$ in both cases). This suggests that the three species naturally divide into two groups $\{\text{Adélie, Chinstrap}\}$ and $\{\text{Gentoo}\}$ with respect to body mass.

5.2.2. Two way ANOVA formulation

We now set up the full two way ANOVA model (with interactions), following equation (1.3.7). Let Y_{ijk} be the body mass (kg) of the k -th penguin of specie i and sex j , for $i = 1, 2, 3$, $j = 1, 2$, $k = 1, \dots, n_{ij}$, and a total of $n = 333$ observations.

- The factor A be the species, with total levels $J = 3$: $i = 1$ (Adélie), $i = 2$ (Chinstrap), $i = 3$ (Gentoo)
- The factor B be the sex, with total levels $K = 2$: $j = 1$ (Female), $j = 2$ (Male).

The corresponding design matrix is $\mathbf{X} \in \mathbb{R}^{333 \times 12}$ and has three units (plus one) such as in example 3.1.4, the species unit U_A with $J = 3$ indicator columns, the sex unit U_B with $K = 2$ indicator columns, and the interaction unit U_γ with $JK = 6$ indicator columns, plus the intercept column unit.

Based on post-hoc analysis, we expect the selected structure to be a cluster within the species unit U_A with $M = 2$ clusters {Adélie, Chinstrap} and {Gentoo} while the sex factor unit U_B remains fully separated with $N = 2$ clusters {Female} and {Male}. This corresponds to the partitions:

$$\mathcal{P}_A = \{\{1, 2\}, \{3\}\}, \quad \mathcal{P}_B = \{\{1\}, \{2\}\},$$

inducing the grid partition

$$\mathcal{R} = \mathcal{P}_A \times \mathcal{P}_B = \{(\{1, 2\}, \{1\}), (\{1, 2\}, \{2\}), (\{3\}, \{1\}), (\{3\}, \{2\})\}$$

for the interaction unit U_γ .

Following equation (3.1.12) the collapsed design matrix is:

$$\mathbf{X}_{I_{\mathcal{R}}} = \left[\mathbf{1}_n \underbrace{\mathbf{x}_{\alpha_1} + \mathbf{x}_{\alpha_2}, \mathbf{x}_{\alpha_3}}_{U_A, M=2} \underbrace{\mathbf{x}_{\beta_1}, \mathbf{x}_{\beta_2}}_{U_B, N=2} \underbrace{\mathbf{x}_{\gamma_{11}} + \mathbf{x}_{\gamma_{21}}, \mathbf{x}_{\gamma_{31}}, \mathbf{x}_{\gamma_{12}} + \mathbf{x}_{\gamma_{22}}, \mathbf{x}_{\gamma_{32}}}_{U_\gamma, MN=4} \right]$$

This collapsed model reduces the full $JK = 6$ cell means to $MN = 4$ distinct cluster means, one for each combination of a species cluster and a sex level. This matrix spans a linear subspace with its statistical dimensions:

$$\mathbb{L}_{I_{\mathcal{R}}} = \text{Col}(\mathbf{X}_{I_{\mathcal{R}}}) \subset \mathbb{R}^{333}, \quad d_{I_{\mathcal{R}}} = \text{rank}(\mathbf{X}_{I_{\mathcal{R}}}) = MN = 4.$$

5.3. Applying the projection framework

5.3.1. Model selection and penalty calibration

The candidate family $\mathcal{I}$ consists of all grid partitions $\mathcal{R} = \mathcal{P}_A \times \mathcal{P}_B$ with $1 \leq M \leq 3$ and $1 \leq N \leq 2$, giving a total of $|\mathcal{I}| = B_3 \times B_2 = 5 \times 2 = 10$ candidate structures, over which the criterion $C(I)$ is minimized. Table 5.2 summarizes some of these ten models with their RSS values; the remaining rows are included in the Appendix ??.

Candidate model	Partition	MN	RSS
1	$\mathcal{P}_A: \{1, 2, 3\} \times \mathcal{P}_B: \{1, 2\}$	$1 \times 1 = 1$	215.260
4	$\mathcal{P}_A: \{1, 3\}, \{2\} \times \mathcal{P}_B: \{1\}, \{2\}$	$2 \times 2 = 4$	155.882
5	$\mathcal{P}_A: \{1, 2\}, \{3\} \times \mathcal{P}_B: \{1, 2\}$	$2 \times 1 = 2$	70.103
9	$\mathcal{P}_A: \{1\}, \{2\}, \{3\} \times \mathcal{P}_B: \{1, 2\}$	$3 \times 1 = 3$	70.069
6	$\mathcal{P}_A: \{1, 2\}, \{3\} \times \mathcal{P}_B: \{1\}, \{2\}$	$2 \times 2 = 4$	32.138
10	$\mathcal{P}_A: \{1\}, \{2\}, \{3\} \times \mathcal{P}_B: \{1\}, \{2\}$	$3 \times 2 = 6$	31.303

Table 5.2.: Selected candidate models with their partition, statistical dimension $d_R = MN$, and RSS (body mass in kg).

We observe that any structure that ignores both factors (model 1) or merges species that are far apart in body mass, such as Adélie with Gentoo (model 4), gives the worst fit. Keeping Gentoo separate from the other species, or keeping the two sexes apart, already improves the fit substantially, and satisfying both at once produces the best fit. The gap between the fully distinguished structure (model 10) and the one that only merges Adélie and Chinstrap (model 6) is very small, which shows that our family contains several nested structures that explain the response almost equally well.

To describe the size of the effects present in the data, we first estimate the noise level. The general framework treats σ^2 as known. With real data, it must be estimated. We use the standard unbiased estimator from the full model residuals:

$$\hat{\sigma}^2 = \frac{\|\mathbf{Y} - \mathbf{P}_{\text{full}}\mathbf{Y}\|^2}{n - \text{rank}(\mathbf{X})} = \frac{\text{RSS}_{\text{full}}}{333 - 6} = \frac{31.303}{327} \approx 0.0957 \quad (5.3.1)$$

where $\mathbf{P}_{\text{full}}$ is the projection onto $\mathbb{L}_{\text{full}} = \text{Col}(\mathbf{X})$ with $\mathbf{X} \in \mathbb{R}^{333 \times 12}$.

To describe the size of the effects present in the data, we compute the signal-to-noise ratios [8, 9] using the six estimated cell means of the classical analysis [25, Descriptive Statistics], converted to kilograms; the step-by-step calculations are provided in the appendix A.6. The relevant contrasts are the two differences in body mass that carry the signal : the within-cluster absolute difference between Adélie and Chinstrap, and the between-cluster absolute difference between the {Adélie, Chinstrap} cluster and {Gentoo}. From equation (5.3.1) we have $\hat{\sigma} = \sqrt{\hat{\sigma}^2} \approx 0.309$ kg and the ratios are:

$$\begin{aligned} \frac{\delta_{\text{within}}}{\hat{\sigma}} &\approx \frac{0.158}{0.309} \approx 0.51 \quad (\text{Female}), & \frac{\delta_{\text{within}}}{\hat{\sigma}} &\approx \frac{0.104}{0.309} \approx 0.34 \quad (\text{Male}), \\ \frac{\delta_{\text{between}}}{\hat{\sigma}} &\approx \frac{1.261}{0.309} \approx 4.08 \quad (\text{Female}), & \frac{\delta_{\text{between}}}{\hat{\sigma}} &\approx \frac{1.475}{0.309} \approx 4.77 \quad (\text{Male}). \end{aligned}$$

The within-cluster ratios are below one, showing that the Adélie and Chinstrap differences are small relative to the noise level, which explains why the full model (model 10) barely improves on the clustered one (model 6) in terms of RSS. In contrast, the

between-cluster ratios are greater than four, indicating a strong separation between the $\{\text{Adelie, Chinstrap}\}$ group and $\{\text{Gentoo}\}$. This large contrast is what the projection estimator captures when Gentoo is kept as a separate block.

We want to minimize the criterion (3.3.4) for this two way ANOVA model over the family of all $|\mathcal{I}| = 10$ candidate structures. Using equation (3.2.8) we get the following:

$$C_{\kappa,\lambda}(I_{\mathcal{R}}) = \text{RSS}_{I_{\mathcal{R}}} + \kappa \hat{\sigma}^2 [MN + \lambda \log(S(J, M) S(K, N))], \quad (5.3.2)$$

We start by tuning (κ, λ) using the same grid and cross-validation method described in section 4.1.2 with $k = 5$ folds. The only difference is that now the κ grid runs up to $\kappa = 200$ rather than $\kappa = 20$. Since here $\hat{\sigma}^2 \approx 0.096$, matching the same effective penalty strength as in the simulation requires $\kappa \approx 20/\hat{\sigma}^2 \approx 200$. To obtain a more stable penalty-selection result we repeat the whole 5-fold procedure over $N = 100$ independent trials, and for each pair (κ, λ) on the grid we evaluate the cross-validation error from (4.1.2). We then average these values and choose the tuned pair following equation (4.1.3)

Interestingly, the result is $(\hat{\kappa}, \hat{\lambda}) = (0.1, 0)$, with $\text{CV}(\hat{\kappa}, \hat{\lambda}) = 0.0982$ which is weakly identified.

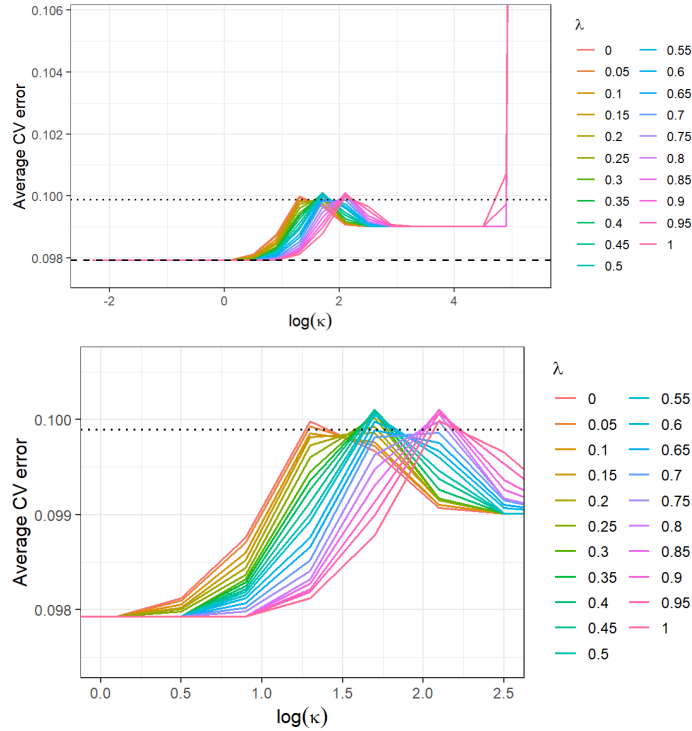


Figure 5.1.: Averaged cross-validation error $\text{CV}(\kappa, \lambda)$, over $N = 100$ runs, against $\log \kappa$, one curve per $\lambda \in [0, 1]$. The dashed line marks the minimum CV and the dotted line lies 2% above it.

As Figure 5.1 shows, for small κ , the cross-validation error is almost flat at its minimum, and the curves for different λ values nearly overlap, which means that several tuning pairs perform equally well. Therefore, the prediction error distinguishes neither a single best κ nor a single best λ : the exact minimum carries little information.

This makes sense; from table 5.2 and the calculated effect sizes we have seen that the family of candidate structures is tiny and that the candidates are nested and very similar. Most of these structures produce almost identical predictions because body mass is driven by one strong, clean signal: Gentoo penguins are much heavier, and males are heavier than females. Any structure that keeps Gentoo separate and keeps the sexes separate therefore captures nearly all the explainable variance.

For such a small κ we expect $\hat{I}_{\mathcal{R}} = \arg \min_{R \in \mathcal{I}} C_{(\hat{\kappa}, \hat{\lambda})}(I_{\mathcal{R}})$ to be the full structure with interactions (model 10).

Because the minimum is not sharply located, we consider the exact minimizer to be a strict choice: a structure with almost the same predictive accuracy but a simpler form can be chosen. Therefore, instead of selecting the exact minimizer of (4.1.3) we use a relaxed cross-validation rule. Among all pairs whose average error lies within 2% of the exact minimum, we select the one with the smaller subspace dimension. Let $CV_{min} = \min_{(\kappa, \lambda)} \overline{CV}(\kappa, \lambda)$. The relaxed problem then is as follows:

$$\begin{aligned} & \min_{(\kappa, \lambda)} d_{I(\kappa, \lambda)} \\ \text{s.t } & CV^*(\kappa, \lambda) \leq 1.02 CV_{min} \end{aligned}$$

Within this band, all the selected structures predict essentially equally well. We therefore choose among them by simplicity: we keep the structure with the smallest dimension MN . Should two of them share the same dimension, the one with the lowest cross-validation error is kept.

As figure 5.2 shows, many tuning values lie within this 2% band (the region from the minimizer up to the dotted line), which confirms that prediction error alone does not separate the minimum-CV model from simpler, near-optimal structures. The rule therefore selects the simplest structure that the data still support within this accuracy margin.

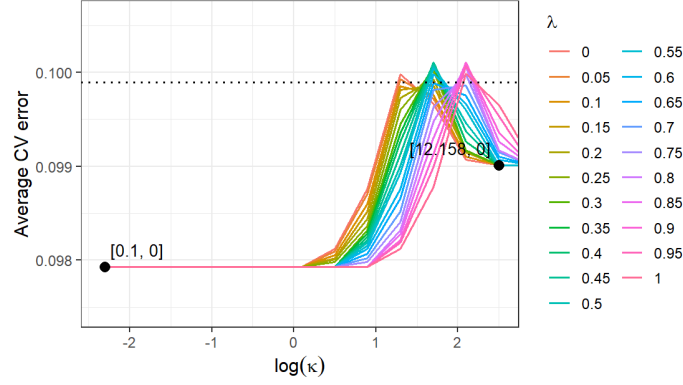


Figure 5.2.: Averaged cross-validation error $CV(\kappa, \lambda)$ over $N = 100$ runs, against $\log(\kappa)$, with one curve per $\lambda \in [0, 1]$. The dotted line marks $1.02 CV_{\min}$, while the black points indicate the exact minimizer $(0.1, 0)$ and the relaxed one $(12.158, 0)$.

Table 5.3 shows the selected structures of the criterion $C_{(\kappa, \lambda)}(I_{\mathcal{R}})$ (5.3.2) for both tuning choices: the pair with the minimum cross-validation error $(0.1, 0)$ and the relaxed one $(12.158, 0)$.

(κ, λ)	CV	Selected structure	MN	RSS
$(0.1, 0)$	0.09817	$\{1\}, \{2\}, \{3\} \times \{1\}, \{2\}$	$3 \times 2 = 6$	31.303
$(12.158, 0)$	0.09923	$\{1, 2\}, \{3\} \times \{1\}, \{2\}$	$2 \times 2 = 4$	32.138

Table 5.3.: Cross-validation error, selected structure, dimension MN , and residual sum of squares for the minimum-CV tuning and for the tuning within 2% of the minimum.

The relaxed cross-validation rule returns $(\kappa, \lambda) = (12.158, 0)$, with $CV = 0.0992$, only 1.07% above the exact minimum $CV_{\min}$. The resulting structure is $\hat{I}_{\mathcal{R}} = \{\{1, 2\}, \{3\}\} \times \{\{1\}, \{2\}\}$, with $MN = 4$: the Adélie and Chinstrap species are merged, while Gentoo and the two sexes remain distinct. The relaxed rule therefore recovers a simpler and interpretable structure, while sacrificing almost nothing in predictive accuracy.

5.3.2. Projection estimation

We estimate the model using the structure chosen by the parsimony rule, not the one at the exact CV minimum. The projection estimator is then $\mathbf{X}_{\hat{I}_{\mathcal{R}}} \hat{\boldsymbol{\beta}}_{\hat{I}_{\mathcal{R}}} = \mathbf{P}_{\hat{I}_{\mathcal{R}}} \mathbf{Y}$, where $\mathbb{L}_{\hat{I}_{\mathcal{R}}} = \text{Col}(\mathbf{X}_{\hat{I}_{\mathcal{R}}})$ is the selected (collapsed) linear subspace of dimension $d_{\hat{I}_{\mathcal{R}}} = MN = 4$. Concretely, for the selected structure $\hat{I}_{\mathcal{R}}$ (model 6), the response vector $\mathbf{Y}$ is projected onto this collapsed subspace $\mathbb{L}_{\hat{I}_{\mathcal{R}}}$. This gives four fitted mean values, one for each combination of the selected species cluster and sex level, and therefore recovers the two-group structure in body mass. Table 5.4 reports the resulting estimates.

Species cluster	Sex	$\mathbf{P}_{\hat{I}_{\mathcal{R}}} \mathbf{Y}$ (kg)
{Adélie, Chinstrap}	Female	3.419
{Adélie, Chinstrap}	Male	4.010
{Gentoo}	Female	4.680
{Gentoo}	Male	5.485

Table 5.4.: Projection estimates $\mathbf{P}_{\hat{I}_{\mathcal{R}}} \mathbf{Y}$ under the selected structure $\hat{I}_{\mathcal{R}}$, with $d_{I_{\mathcal{R}}} = 4$ distinct cell means (body mass in kg).

The four estimated cell means reflect the collapsed structure $\hat{I}_{\mathcal{R}}$ directly: Adélie and Chinstrap share a single estimated mean per sex level (3.419 kg for females and 4.010 kg for males), while Gentoo retains its own estimates (4.680 kg and 5.485 kg, respectively). These values are consistent with the full-model mean of the classical analysis [25, Descriptive Statistics]. The collapsed female estimate of 3.419 kg lies between the individual Adélie (3.369 kg) and Chinstrap (3.527 kg) female means and similarly for males. The Gentoo estimates are unchanged, as expected for a species kept as a separate block.

6. Discussion

6.1. Contribution of this thesis and main findings

This thesis is based on the general framework for projection structures developed by E.Belitser and N.Nurushev [11] which works with abstract families of candidate linear subspaces and a majorant penalty defined on them. That framework is developed on the theoretical side for high-dimensional statistics: it constructs a data-dependent measure and derives theoretical guarantees. The present work takes a specific part of this framework, the frequentist formulation, which takes the form of a penalized criterion, and studies it empirically. Candidate families and majorants are explicitly defined for specific ANCOVA models, and the resulting criterion is examined through simulations and a real data application. The main contributions with their findings are the following.

Contribution 1 We make the two structures of interest, sparsity and clustering, concrete for the ANCOVA models by describing them through the columns of the design matrix. This description is based on the unit notion, a set of columns that conceptually belong together. Unit sparsity is then an active set of units, clustering within a unit is a partition of its column indices, and clustering within an interaction unit is the Cartesian product of two partitions. For each structure I , an explicit design matrix $\mathbf{X}_I$ is defined and hence its linear subspace $\mathbb{L}_I = \text{Col}(\mathbf{X}_I)$ with statistical dimension $d_I = \text{rank}(\mathbf{X}_I)$.

Main finding 1 The mapping matrix $T \in \{0, 1\}^{J \times M}$ used to define the clustering matrix makes precise the observation that the sparsity of multi-column units is recovered from the clustering within that unit by restricting to the two extreme partitions. When $T = \mathbf{I}_{J \times J}$ ($M = J$) no indices are merged and the unit is fully present; when $T = \mathbf{1}_J$ ($M = 1$) all indices collapse into the intercept and the unit disappears.

Contribution 2 For each structure I , we give a closed-form expression for the combinatorial term $\ell(I)$ in terms of combinatorial quantities: the Stirling number $S(J, M)$ for clustering, the binomial coefficient $\binom{q}{s}$ for sparsity, and their products for composite structures. Together with the statistical dimension of the linear subspaces from contribution 1, these terms give a closed form for the majorant $\rho(I) = d_I + \log \ell(I)$. The total sum of these combinatorial quantities B_J and 2^q gives the family size of all admissible structures $|\mathcal{I}|$.

Main finding 2 When two structures are combined, for instance, clustering with sparsity or clustering with clustering, the combinatorial term of the composite structure factorizes into the product of the two individual combinatorial terms. In the two-way ANOVA model, this makes the additive and interaction structures directly comparable: they share the same combinatorial term and differ only in statistical dimension. A short calculation then shows that the penalized criterion prefers the interaction structure only when it reduces RSS by more than $\kappa\sigma^2(M-1)(N-1)$.

Contribution 3 In the considered low-dimensional ANOVA settings, the theoretical majorant overpenalises any intermediate clustering $1 < M < J$ because such structures have larger combinatorial terms $\ell(I)$ than the extreme ones with $M = J$. We identify this limitation and propose an adjusted majorant $\rho_\lambda(I) = d_I + \lambda \log \ell(I)$, which adds some weight to the combinatorial term. This is a practical finite-sample calibration with empirical rather than theoretical support. This adjusted majorant enters the penalized criterion $C_{\kappa,\lambda}(I) = RSS_I + \kappa\sigma^2[d_I + \lambda \log \ell(I)]$, which now has two tuning parameters: the overall penalty strength $\kappa > 0$ and the combinatorial weight $\lambda \in [0, 1]$

Main finding 3 The extreme case of $\lambda = 0$ gives an interesting connection. At this value the majorant reduces to a dimension-based penalty and we get $C_{\kappa,0} = RSS_I + \kappa\sigma^2 d_I$ which is in the same spirit as AIC and Mallows' C_p for low-dimensional Gaussian linear models. At $\lambda = 1$ the full theoretical majorant is kept, and using $S(J, M) \sim M^J/M!$ as the number of levels grows, it can recover a two-cluster structure in the limit.

Contribution 4 We study the behavior of the penalized criterion $C_{\kappa,\lambda}(I)$, from contribution 3, empirically, through simulations and a real data application, across all considered models. The two tuning parameters (κ, λ) are selected jointly: cross-validation provides the data-driven choice, while recovery-rate tables and heatmaps describe how the (κ, λ) grid governs which structure is recovered.

Main finding 4 The two parameters κ and λ act differently. The weight λ strongly restricts the clustering structures but barely affects sparsity, a contrast that follows directly from the different combinatorial terms involved: the clustering cost peaks at the true structure, while the sparsity cost cannot separate the true structure from its near competitors.

Contribution 5 For small families of nearly equivalent nested models, where the cross-validation error is almost flat, we introduce a relaxed selection rule: among all tunings whose error lies within a small band of the minimum, we keep the structure of the smallest dimension.

Main finding 5. The relaxed rule maintains a simpler and more interpretable structure with almost no loss in predictive error. On the Palmer penguins data it selects a four-cell structure instead of the full six-cell one, at a cross-validation error only slightly above the minimum.

6.2. Limitations and possible extensions

The projection framework we use is developed for high-dimensional statistics and comes with theoretical recovery guaranties for a very rich family $\mathcal{I}$ of candidate structures. This thesis did not work in that setting. We only treated low-dimensional models, one way ANOVA, two way ANOVA, and one factor ANCOVA with q covariates without interactions, and we found the minimizer $\hat{I} = \arg \min_{I \in \mathcal{I}} C_{\kappa, \lambda}(I)$ by an exhaustive search over the whole family. This brute-force search is only feasible because $\mathcal{I}$ is small here: for our models $|\mathcal{I}|$ is at most B_J , 2^q , or $B_J \cdot 2^q$ which are kept small in our study ($J = 5$, $q = 11$). As soon as these numbers grow, the family $\mathcal{I}$ becomes too large to enumerate, and evaluating $C_{\kappa, \lambda}(I)$ in every candidate structure is no longer possible.

A natural next step is to apply the framework to genuinely high-dimensional ANCOVA models, with more factors, more covariates, and factor-covariate interactions, and to replace the exhaustive search with a dedicated minimization algorithm. Because a full ANCOVA structure is a pair or product of two structures $I_{\mathcal{P}}$ and $I^{(S)}$, the two parts suggest two different tools. The unit sparsity part $I^{(S)}$ selects an active set of units and could be handled by a branch-and-bound scheme that prunes subsets of units without visiting all of 2^q . The clustering part $I_{\mathcal{P}}$ searches the partitions of a unit, so a dynamic-programming recursion over blocks could reuse the combinatorial structure of $S(J, M)$ instead of listing all B_J partitions. Combining the two in one algorithm would let the criterion $C_{\kappa, \lambda}(I)$ be minimized in high dimensions without evaluating every candidate.

Another limitation is that the candidate family $\mathcal{I}$ and its associated majorant were constructed separately for pre-specified ANCOVA families: one way ANOVA, two way ANOVA with and without interactions, and ANCOVA with one factor and q covariates without interactions. The criterion therefore selects the best structure within the chosen family but does not select among the families themselves. In the Palmer penguins application, for example, the family of full two way ANOVA structures with interactions was used because the preliminary classical analysis had already indicated the presence of an interaction. The projection criterion then selected the clustering structure within this interaction model but was not asked to decide whether the interaction itself should be kept.

A possible extension is to define an enlarged family $\mathcal{I}_{\text{all}}$ that contains both additive and interaction structures, together with their nested clustering and sparsity cases, and to construct a single majorant that satisfies condition (C2) in this entire family. Minimizing

the same criterion in $\mathcal{I}_{\text{all}}$ would then jointly determine which family to choose, and which structure, that is, which covariates are active and which factor levels should be clustered.

Finally, recovery rates in this thesis are measured at a single signal level. In the one way ANOVA the level means are separated by an effective distance of about 6.7 standard errors, and in the ANCOVA the weakest active covariate has an effective separation from zero of about 15 standard errors. Both signals are strong in relation to the noise. This isolates the effect of the tuning (κ, λ) , so that the recovery rates reflect the penalty alone and not the difficulty of detection. The selected (κ, λ) values for each simulation, are chosen using recovery rates and cross-validation only at this strong signal. It is not known whether the same (κ, λ) still gives the highest recovery rate or the best prediction of new observations when the signal is weaker.

A further question is how the strength of the signal affects the selection. Holding the design fixed, one would let the cluster separation and the covariate signal range from weak to strong, and record how often the true structure is selected at each level. This would show at which signal strength the selection becomes reliable, and whether the selected (κ, λ) stays the same or changes as the signal varies.

6.3. The benefits of applying the general framework for projection structures to ANCOVA models

The projection framework turns model selection in ANCOVA into a geometric choice between linear subspaces, rather than a choice between individual coefficients. Ordinary least squares already fits a fixed linear model by projecting $\mathbf{Y}$ onto the column space of its design matrix. The projection framework takes this one step further and asks which candidate subspace should be used for the projection. This is useful because the comparison is made directly at the level of the fitted object in $\mathbb{R}^n$, rather than through a sequence of separate coefficient tests or coefficient restrictions.

In addition, there is no need for additional identifiability constraints or reparameterization. These usually appear in ANCOVA models due to the dummy-variable encoding, in order to make their design matrix $\mathbf{X}$ of full rank and hence make possible the unique estimation of the coefficients $\hat{\boldsymbol{\beta}}$. In the projection framework, the quantity of interest is the fitted mean vector $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{P}_I\mathbf{Y}$, which is unique and well-defined for every admissible structure I as an orthogonal projection[6].

Furthermore, different structural questions can all be addressed using the same criterion $C_\kappa(I) = \text{RSS}_I + \kappa\sigma^2\rho(I)$. Its minimization answers whether a covariate should be included, whether a categorical factor should be kept, and whether its levels should be merged. A further benefit concerns the nature of the model selection procedure itself:

this is carried out in a single step over the entire discrete family $\mathcal{I}$, delivering both the selected structure $\hat{I}$ and the mean estimator $P_{\hat{I}}\mathbf{Y}$ simultaneously.

Finally, any new ANCOVA structure can be included in the framework. This is useful because the same machinery can be reused: define the subspace and its statistical dimension, take its projection, compute the combinatorial term, and compare it by the same criterion. The framework is particularly attractive because it was originally developed for broad high-dimensional settings, while this thesis applies it in a concrete low-dimensional context. This leaves a natural route for extending the thesis to richer ANCOVA models with many covariates, more factors, or more complex interaction structures.

A. Appendix

A.1. Additional simulation results for section 4.1.1

Panel A: $\lambda = 0, \dots, 0.50$

κ	λ										
	0	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
0.100	0.055	0.050	0.045	0.035	0.035	0.025	0.025	0.020	0.015	0.010	0.000
0.156	0.105	0.100	0.085	0.085	0.085	0.075	0.070	0.060	0.050	0.030	0.010
0.242	0.155	0.150	0.140	0.140	0.130	0.120	0.105	0.095	0.085	0.075	0.035
0.376	0.220	0.215	0.205	0.205	0.185	0.170	0.165	0.150	0.130	0.100	0.075
0.585	0.310	0.300	0.295	0.280	0.265	0.250	0.235	0.225	0.190	0.165	0.115
0.909	0.420	0.405	0.385	0.365	0.350	0.330	0.315	0.295	0.275	0.240	0.185
1.414	0.575	0.555	0.545	0.520	0.500	0.480	0.445	0.400	0.380	0.310	0.260
2.199	0.755	0.735	0.705	0.700	0.670	0.660	0.615	0.585	0.520	0.440	0.330
3.420	0.875	0.865	0.855	0.845	0.830	0.810	0.785	0.750	0.705	0.620	0.465
5.318	0.950	0.950	0.940	0.920	0.910	0.895	0.875	0.860	0.840	0.800	0.650
8.270	0.980	0.975	0.970	0.965	0.965	0.965	0.960	0.935	0.900	0.870	0.805
12.861	0.935	0.925	0.920	0.920	0.905	0.905	0.900	0.895	0.885	0.865	0.800
20.000	0.815	0.795	0.750	0.735	0.735	0.710	0.695	0.675	0.655	0.645	0.615

Panel B: $\lambda = 0.55, \dots, 1$

κ	λ									
	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95	1
0.100	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.156	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.242	0.010	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.376	0.030	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.585	0.075	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.909	0.100	0.010	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1.414	0.165	0.040	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2.199	0.240	0.075	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3.420	0.315	0.115	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5.318	0.440	0.185	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
8.270	0.610	0.260	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
12.861	0.705	0.295	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
20.000	0.550	0.265	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table A.1.: Complete recovery rates of $I_{\mathcal{P}_{\text{true}}}$ among the 52 admissible candidate structures, over $N_{\text{sim}} = 200$ simulations, for a range of values of (κ, λ) .

A.2. Additional simulation results for section 4.1.2

Panel A: $\lambda = 0, \dots, 0.50$

κ	λ										
	0	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
0.100	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075
0.156	1.074	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075
0.242	1.074	1.074	1.074	1.074	1.074	1.075	1.075	1.075	1.075	1.075	1.075
0.376	1.073	1.073	1.073	1.073	1.074	1.074	1.074	1.075	1.075	1.075	1.075
0.585	1.071	1.071	1.071	1.071	1.072	1.073	1.073	1.074	1.074	1.075	1.075
0.909	1.070	1.070	1.070	1.070	1.070	1.071	1.071	1.072	1.073	1.074	1.074
1.414	1.066	1.066	1.066	1.067	1.068	1.068	1.069	1.070	1.071	1.072	1.073
2.199	1.063	1.063	1.063	1.064	1.065	1.065	1.066	1.067	1.068	1.069	1.072
3.420	1.062	1.062	1.061	1.061	1.062	1.062	1.063	1.064	1.065	1.066	1.068
5.318	1.059	1.059	1.060	1.060	1.060	1.061	1.062	1.063	1.063	1.065	1.067
8.270	1.062	1.063	1.063	1.063	1.064	1.065	1.066	1.066	1.067	1.068	1.071
12.861	1.084	1.086	1.090	1.092	1.096	1.099	1.103	1.108	1.114	1.118	1.121
20.000	1.182	1.190	1.204	1.215	1.223	1.234	1.240	1.257	1.265	1.280	1.295

Panel B: $\lambda = 0.55, \dots, 1$

κ	λ									
	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95	1
0.100	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075
0.156	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075
0.242	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075
0.376	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075
0.585	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075
0.909	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075
1.414	1.074	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075
2.199	1.073	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075
3.420	1.072	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075	1.075
5.318	1.071	1.075	1.076	1.075	1.075	1.075	1.075	1.075	1.075	1.075
8.270	1.074	1.080	1.081	1.078	1.076	1.076	1.075	1.075	1.075	1.075
12.861	1.125	1.129	1.112	1.098	1.088	1.083	1.080	1.078	1.077	1.077
20.000	1.304	1.307	1.277	1.242	1.215	1.200	1.186	1.178	1.175	1.174

Table A.2.: Averaged cross-validation errors over $N_{\text{sim}} = 200$ runs for all combinations of (κ, λ) .

A.3. Additional simulation results for section 4.2

J	M_{true}	ρ_{true}	ρ_{full}	Recovery (ρ)
5	2	4.71	5	0.345
5	3	6.22	5	0.000
5	4	6.30	5	0.000
10	2	8.24	10	0.690
10	5	15.66	10	0.000
10	7	15.68	10	0.000
10	9	12.81	10	0.000
20	2	15.17	20	0.865
20	5	32.34	20	0.000
20	10	39.41	20	0.000
20	15	34.93	20	0.000
20	19	24.25	20	0.000
40	2	29.03	40	0.935
40	10	86.85	40	0.000
40	20	94.17	40	0.000
40	30	78.10	40	0.000
40	39	45.66	40	0.000
60	2	42.90	60	0.995
60	15	149.34	60	0.000
60	30	154.29	60	0.000
60	45	124.02	60	0.000
60	59	66.48	60	0.000
80	2	56.76	80	1.000
80	20	216.98	80	0.000
80	40	217.88	80	0.000
80	60	171.69	80	0.000
80	79	87.06	80	0.000
90	2	63.69	90	1.000
90	22	251.38	90	0.000
90	45	250.67	90	0.000
90	67	197.72	90	0.000
90	89	97.30	90	0.000

Table A.3.: Complete values of the theoretical majorant for the true cluster M_{true} and the full model and recovery rates, over $N_{\text{sim}} = 200$ runs, varying both J and M_{true} at fixed $\kappa = 5.32$

A.4. Additional simulation results for section 4.3.2

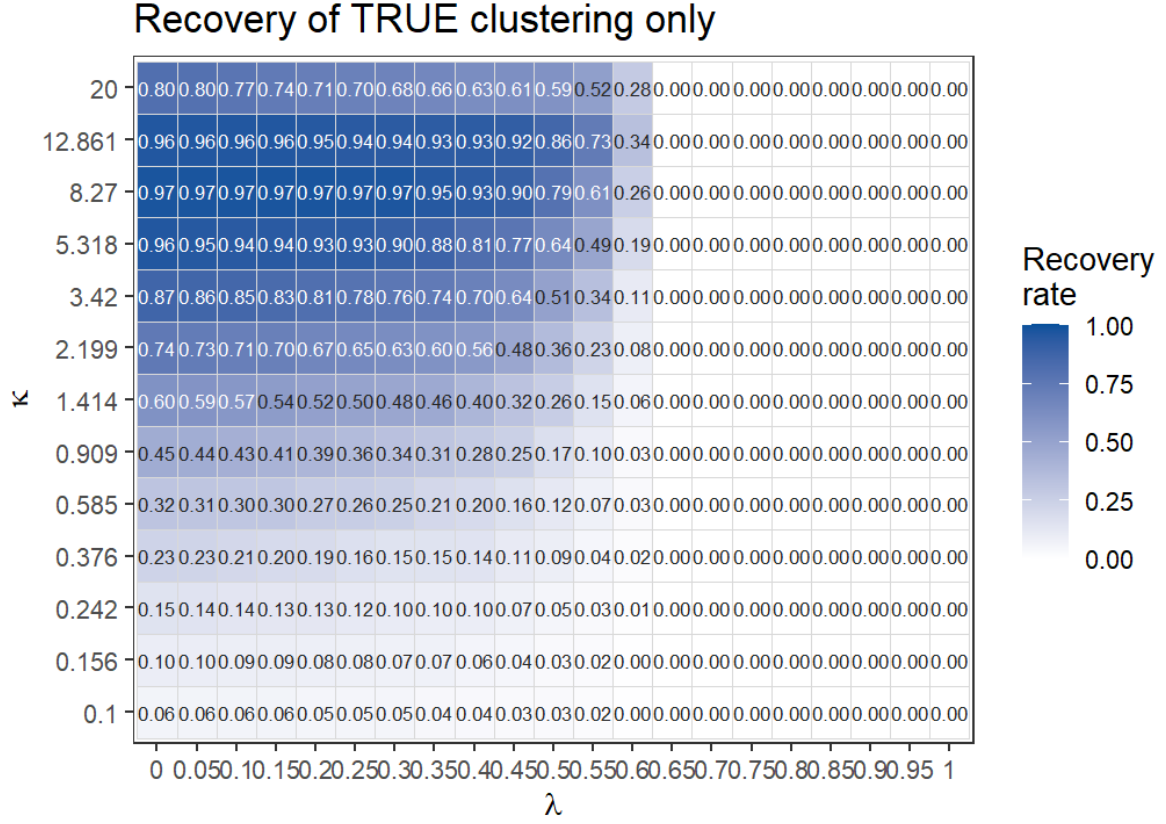


Figure A.1.: Recovery rates of the $I_{\mathcal{P}_{true}}$, among all admissible candidate structures, over $N_{sim} = 200$ simulations, for the full range of values of (κ, λ) .

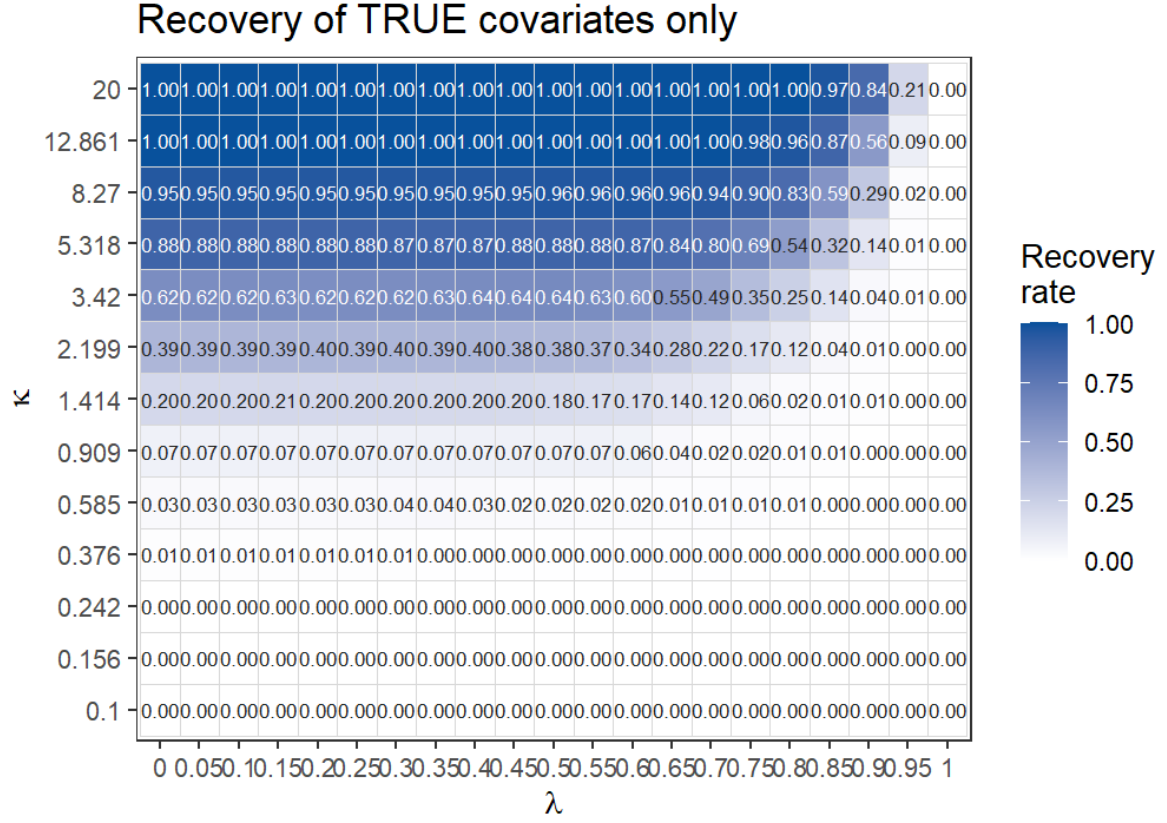


Figure A.2.: Recovery rates of the $I^{(S_{true})}$, among all admissible candidate structures, over $N_{sim} = 200$ simulations, for the full range of values of (κ, λ) .

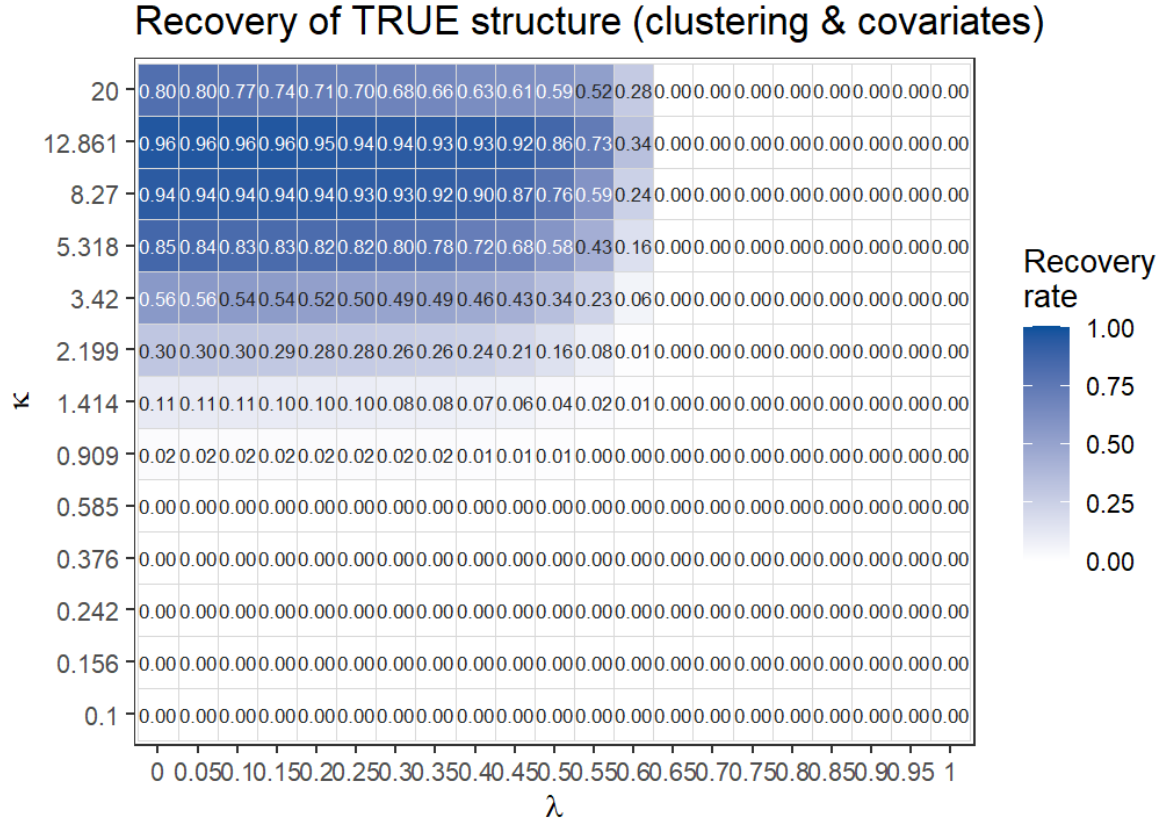


Figure A.3.: Recovery rates of the $I_{\mathcal{P}_{true}}$ and $I^{(S_{true})}$ simultaneously, among all admissible candidate structures, over $N_{sim} = 200$ simulations, for the full range of values of (κ, λ) .

A.5. Complete candidate model table for 6.3.1

Table 5.2 summarizes the selected candidate models with their RSS values. The complete set of ten candidate models is included here in table, A.4.

Candidate model	Partition	MN	RSS
1	$\mathcal{P}_A: \{1, 2, 3\} \times \mathcal{P}_B: \{1, 2\}$	$1 \times 1 = 1$	215.260
2	$\mathcal{P}_A: \{1, 2, 3\} \times \mathcal{P}_B: \{1\}, \{2\}$	$1 \times 2 = 2$	176.381
3	$\mathcal{P}_A: \{1, 3\}, \{2\} \times \mathcal{P}_B: \{1, 2\}$	$2 \times 1 = 2$	196.064
4	$\mathcal{P}_A: \{1, 3\}, \{2\} \times \mathcal{P}_B: \{1\}, \{2\}$	$2 \times 2 = 4$	155.882
5	$\mathcal{P}_A: \{1, 2\}, \{3\} \times \mathcal{P}_B: \{1, 2\}$	$2 \times 1 = 2$	70.103
6	$\mathcal{P}_A: \{1, 2\}, \{3\} \times \mathcal{P}_B: \{1\}, \{2\}$	$2 \times 2 = 4$	32.138
7	$\mathcal{P}_A: \{2, 3\}, \{1\} \times \mathcal{P}_B: \{1, 2\}$	$2 \times 1 = 2$	150.030
8	$\mathcal{P}_A: \{2, 3\}, \{1\} \times \mathcal{P}_B: \{1\}, \{2\}$	$2 \times 2 = 4$	111.946
9	$\mathcal{P}_A: \{1\}, \{2\}, \{3\} \times \mathcal{P}_B: \{1, 2\}$	$3 \times 1 = 3$	70.069
10	$\mathcal{P}_A: \{1\}, \{2\}, \{3\} \times \mathcal{P}_B: \{1\}, \{2\}$	$3 \times 2 = 6$	31.303

Table A.4.: Complete list of candidate models with their partitions, statistical dimension $d_R = MN$, and RSS (body mass in kg).

A.6. Signal-to-noise ratio computations for the Palmer penguins data study

Cell means in kg. We first report the six estimated cell means from the classical two way ANOVA analysis of [25, Descriptive Statistics], converted into kilograms. We will use these six cell means as fixed effects for the signal computations.

Species	Female (kg)	Male (kg)
Adélie	3.369	4.043
Chinstrap	3.527	3.939
Gentoo	4.680	5.485

Table A.5.: Estimated cell means in kg by species and sex

Within-cluster signal δ_{within} . Based on the post-hoc analysis from [25] we suspect species 1 and 2 {Adélie, Chinstrap} form one cluster, while species 3 {Gentoo} stands alone. The within-cluster is the separation between Adélie and Chinstrap within each sex.

$$\delta_{\text{within,female}} = |\mu_{\text{Adélie,female}} - \mu_{\text{Chinstrap,female}}| = |3.369 - 3.527| = 0.158 \text{ kg}, \quad (\text{A.1})$$

$$\delta_{\text{within,male}} = |\mu_{\text{Adélie,male}} - \mu_{\text{Chinstrap,male}}| = |4.043 - 3.939| = 0.104 \text{ kg}. \quad (\text{A.2})$$

These give the within-cluster ratios quoted in the main text for the female and male directions, respectively.

Between-cluster signal δ_{between} . The between-cluster contrast compares {Gentoo} to the merged {Adélie, Chinstrap} cluster, within each sex; this is the direction that the true projection structure is supposed to separate. For each sex, we first average Adélie and Chinstrap to get their cluster mean and then find its distance from the Gentoo:

$$\bar{\mu}_{\{\text{Adélie, Chinstrap}\},\text{female}} = \frac{\mu_{\text{Adélie,female}} + \mu_{\text{Chinstrap,female}}}{2} = \frac{3.369 + 3.527}{2} = 3.448 \text{ kg}, \quad (\text{A.3})$$

$$\bar{\mu}_{\{\text{Adélie, Chinstrap}\},\text{male}} = \frac{\mu_{\text{Adélie,male}} + \mu_{\text{Chinstrap,male}}}{2} = \frac{4.043 + 3.939}{2} = 3.991 \text{ kg}. \quad (\text{A.4})$$

The between-cluster signals are then

$$\delta_{\text{between,female}} = |\mu_{\text{Gentoo,female}} - \bar{\mu}_{\{\text{Adélie, Chinstrap}\},\text{female}}| = |4.680 - 3.448| = 1.232 \text{ kg}, \quad (\text{A.5})$$

$$\delta_{\text{between,male}} = |\mu_{\text{Gentoo,male}} - \bar{\mu}_{\{\text{Adélie, Chinstrap}\},\text{male}}| = |5.485 - 3.991| = 1.494 \text{ kg}. \quad (\text{A.6})$$

These give the between-cluster signal-to-noise ratios.

A.7. Code Availability

The R-scripts developed for the simulation and the real data study can be found at: <https://posit.cloud/content/12550890>.¹ The scripts are written in a way that each can run and give results individual of the others; except the package installation that is needed for all.

R-script file #	R-script file name	Content
1	Packages and libraries	Installs the required packages and libraries for both simulation and real data study.
2	One way-tuning with recovery rates	Sets up the simulation for 4.1, and tunes (κ, λ) via recovery rates for section 4.1.1
3	One way-tuning via CV	Sets up the simulation for 4.1, and tunes (κ, λ) using cross validation for section 4.1.2
4	One-way selection using both tunings	Sets up the simulation for 4.1, and minimizes the criterion using both tuned pairs (κ, λ) from the recovery rates and CV for section 4.1.3
5	Search over J and M	Sets up the simulation for 4.1, and searches over J and M for section 4.2.
6	ANCOVA simulation	Sets up the simulation for 4.3.1, and tunes (κ, λ) via recovery rates for section 4.3.2.
5	PalmerPenguins	Real-data study using the Palmer penguins data set for 5.

Table A.6.: R scripts used for the simulation study and real data analysis.

¹The reader needs to log in with a Google or GitHub account to access the project.

Bibliography

- [1] S. Banerjee and A. Roy. *Linear Algebra and Matrix Analysis for Statistics*. Chapman & Hall/CRC Texts in Statistical Science. CRC Press, Taylor & Francis Group, Boca Raton, FL, 2014. ISBN 978-1-4822-4824-1. Chapters used: 1.4, 4.6(p. 97),6.2 ,6.4 ,8.3 ,16.8 (p. 552).
- [2] G. Strang. *Linear Algebra and Its Applications*, Fourth Edition. Thomson, Brooks/-Cole, Belmont, CA, 2006. Sections used: 3.3
- [3] R. A. Brualdi. *Introductory Combinatorics*, Fifth Edition. Pearson Education / Prentice Hall, 2010. ISBN 978-0-13-602040-0. Chapters used: 2.1 (p. 27–29), 8.2 (p. 283–288).
- [4] R. Grassl and O. Levin. *Exploring Combinatorial Mathematics*, First Edition. University of Northern Colorado, 2019. Available online at <https://discrete.openmathbooks.org/ecm/>. Chapters used: 2.3 (p. 86–97),
- [5] P.R.Halmos. *Naive Set Theory*. D. Van Nostrand Company, Inc., Princeton, NJ, 1960.
- [6] G.A.F.Seber and A.J.Lee. *Linear Regression Analysis*, Second Edition. Wiley-Interscience, 2003. ISBN 0-471-41540-5. Chapters used: 1.1–1.3, 3.1–3.3, 8.1–8.3,8.8, 12
- [7] M.H.Kutner, C.J.Nachtsheim, J.Neter, and W.Li. *Applied Linear Statistical Models*, Fifth Edition. McGraw-Hill/Irwin, Boston, 2005. ISBN 978-0-07-310874-2. Chapters used: 1.6, 2.7–2.8 , 5.5–5.11, 6.1–6.2,7.2–7.3, 16, 17, 19, 22
- [8] D. C. Montgomery. *Design and Analysis of Experiments*, Eighth Edition. John Wiley & Sons, 2013. ISBN: 978-1-118-14692-7. Chapters used: 3.5.6–3.5.8,3.8–3.11, 10.
- [9] J.Cohen. *Statistical Power Analysis for the Behavioral Sciences*, Second Edition. Lawrence Erlbaum Associates, Publishers, Hillsdale, NJ, 1988. ISBN 0-8058-0283-5. Chapters used: 2, 8
- [10] G.James, D.Witten, T.Hastie, and R.Tibshirani. *An Introduction to Statistical Learning: with Applications in R*. Springer Texts in Statistics. Springer, New York, 2013. ISBN 978-1-4614-7137-0. DOI: 10.1007/978-1-4614-7138-7. Chapters used: 5, 6.

- [11] Belitser, E. and Nurushev, N. (2019). General framework for projection structures. *arXiv preprint arXiv:1904.01003v2 [math.ST]*. Available at <https://arxiv.org/abs/1904.01003>.
- [12] Y. She. Sparse regression with exact clustering. *Electronic Journal of Statistics*, 4:1055–1096, 2010. DOI: 10.1214/10-EJS578.
- [13] H. D. Bondell and B. J. Reich. Simultaneous factor selection and collapsing levels in ANOVA. *Biometrics*, 65(1):169–177, 2009. DOI: 10.1111/j.1541-0420.2008.01061.x.
- [14] J.B.Post and H.D.Bondell. Factor selection and structural identification in the interaction ANOVA model. *Biometrics*, 69(1):70–79, 2013. DOI: 10.1111/j.1541-0420.2012.01810.x.
- [15] J.Gertheiss and G.Tutz. Sparse modeling of categorical explanatory variables. *The Annals of Applied Statistics*, 4(4):2150–2180, 2010. DOI: 10.1214/10-AOAS355.
- [16] A.Maj-Kańska, P.Pokarowski, and A.Prochenka. Delete or merge regressors for linear model selection. *Electronic Journal of Statistics*, 9(2):1749–1778, 2015. DOI: 10.1214/15-EJS1050.
- [17] K.Behdin, R.Benbaki, P.Radchenko, and R.Mazumder. Modeling with categorical features via exact fusion and sparsity regularisation. *arXiv preprint arXiv:2603.25854v1 [stat.ME]*, 2026. Available at <https://arxiv.org/abs/2603.25854>.
- [18] S.Nowakowski, P.Pokarowski, W.Rejchel, and A.Softys. Improving Group Lasso for high-dimensional categorical data. 2023. Available online at <https://github.com/SzymonNowakowski/ICCS-2023>.
- [19] J.Schepers, H.-H.Bock, and I.Van Mechelen. Maximal interaction two-mode clustering. *Journal of Classification*, 34:49–75, 2017. DOI: 10.1007/s00357-017-9226-x.
- [20] H. Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974. Available at: https://eclass.uoa.gr/modules/document/file.php/MATH452/%CE%86%CF%81%CE%B8%CF%81%CE%B1/Akaike_1974.pdf.
- [21] C. L. Mallows. Some comments on C_p . *Technometrics*, 15(4):661–675, 1973. Available at: <https://sites.stat.washington.edu/courses/stat527/s13/readings/technometrics1973.pdf>.
- [22] A.M.Horst, A.P.Hill, and K.B.Gorman. *Palmer Archipelago Penguins Data in the palmerpenguins R Package: An Alternative to Anderson’s Irises*. The R Journal, 14(1):244–254, 2022. Available at: https://www.researchgate.net/publication/362573623_Palmer_Archipelago_Penguins_Data_in_the_palmerpenguins_R_Package_-_An_Alternative_to_Anderson's_Irises

- [23] N. J. Higham. What is a submatrix? *Higham Blog*, October 11, 2023. Available online at <https://nhigham.com/2023/10/11/what-is-a-submatrix/> Accessed: 3 May 2026.
- [24] ProofWiki. Definition: Cartesian Product/Family of Sets. Last modified 13 October 2023. Available online at https://proofwiki.org/wiki/Definition:Cartesian_Product/Family_of_Sets. Accessed: 3 May 2026.
- [25] A.Soetewey. two way ANOVA in R. *Stats and R*, 19 June 2023. Available online at <https://statsandr.com/blog/two-way-anova-in-r/#data> Accessed: 10 May 2026.
- [26] K.B.Gorman, T.D.Williams, and W.R.Fraser. Ecological sexual dimorphism and environmental variability within a community of Antarctic penguins (genus *Pygoscelis*). *PLoS ONE*, 9(3):e90081, 2014. DOI: 10.1371/journal.pone.0090081.
- [27] A.M.Horst, A.P.Hill, and K.B.Gorman. *palmerpenguins: Palmer Archipelago (Antarctica) penguin data*. R package version 0.1.0, 2020. DOI: 10.5281/zenodo.3960218. Available online at <https://allisonhorst.github.io/palmerpenguins/>.